

NEIGHBOURGUARD-SMART HAZARD REPORTING AND RESPONSE SYSTEM

A PROJECT REPORT

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ABSTRACT

NeighbourGuard is a web-based smart hazard reporting and response system designed to improve community safety through efficient identification, validation, and management of hazards. The system enables users to report incidents by providing textual descriptions, uploading images, recording audio, and selecting locations through an interactive map interface, ensuring accurate and comprehensive reporting. To enhance the reliability of reported incidents, the system incorporates an artificial intelligence-based validation mechanism that combines a trust scoring approach with a Convolutional Neural Network (CNN) model, specifically `fire_detection_model.h5` and `stray_animal.h5`, to analyze and verify uploaded images for hazards such as fire, flood, and related emergencies. Only reports that achieve a trust score above 60% are considered valid and are stored in the reports table of the MySQL database, thereby reducing false or misleading information.

Validated reports are stored in a centralized database, allowing authorities and administrators to monitor, analyze, and respond effectively. In addition, the system includes an emergency alert feature that enables users to trigger critical alerts, which are processed using Natural Language Processing (NLP) techniques to extract key information from user descriptions and automatically categorize the type of emergency. These categorized alerts are displayed on the admin dashboard for rapid monitoring and response.

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LIST OF ABBREVIATIONS

AI	- ARTIFICIAL INTELLIGENCE
API	- APPLICATION PROGRAMMING INTERFACE
CNN	- CONVOLUTIONAL NEURAL NETWORK
CPU	- CENTRAL PROCESSING UNIT
CSS	- CASCADING STYLE SHEETS
curl	- CLIENT URL
DB	- DATABASE
DFD	- DATA FLOW DIAGRAM
ER	- ENTITY RELATIONSHIP
FK	- FOREIGN KEY
Flask	- PYTHON WEB MICRO FRAMEWORK
GPS	- GLOBAL POSITIONING SYSTEM
HTML	- HYPER TEXT MARKUP LANGUAGE
HTTP	- HYPER TEXT TRANSFER PROTOCOL
HTTPS	- HYPER TEXT TRANSFER PROTOCOL SECURE
IDE	- INTEGRATED DEVELOPMENT ENVIRONMENT
IoT	- INTERNET OF THINGS
JS	- JAVASCRIPT
JSON	- JAVASCRIPT OBJECT NOTATION
ML	- MACHINE LEARNING
MySQL	- MY STRUCTURED QUERY LANGUAGE
NLTK	- NATURAL LANGUAGE TOOLKIT
NLP	- NATURAL LANGUAGE PROCESSING
OTP	- ONE TIME PASSWORD
PDO	- PHP DATA OBJECTS
PHP	- HYPERTEXT PREPROCESSOR
PK	- PRIMARY KEY
RAM	- RANDOM ACCESS MEMORY

RBAC	- ROLE BASED ACCESS CONTROL
REST	- REPRESENTATIONAL STATE TRANSFER
RF	- RANDOM FOREST
ROM	- READ ONLY MEMORY
SMTP	- SIMPLE MAIL TRANSFER PROTOCOL
SQL	- STRUCTURED QUERY LANGUAGE
SSL	- SECURE SOCKET LAYER
TF-IDF	- TERM FREQUENCY INVERSE DOCUMENT FREQUENCY
UI	- USER INTERFACE
UML	- UNIFIED MODELLING LANGUAGE
URL	- UNIFORM RESOURCE LOCATOR
UX	- USER EXPERIENCE
WWW	- WORLD WIDE WEB

CHAPTER 1

INTRODUCTION

1.1 ABOUT THE PROJECT

In an era of rapid technological advancement, where digital solutions are transforming the way societies function, one critical area that demands immediate attention is community safety and hazard management.

The delay in emergency reporting and response has severe consequences. A fire that goes unreported for even a few minutes can spread beyond control. A flood that is not communicated to authorities promptly can trap residents in affected areas. A road accident that is not immediately reported can result in victims not receiving timely medical attention.

NeighbourGuard: Smart Hazard Reporting and Response System is a modern web-based solution that bridges this critical gap by leveraging advanced web technologies and artificial intelligence. The system empowers individuals within a community to actively participate in safety management by reporting hazards in real time. Through a simple and user-friendly web interface, individuals can submit detailed reports using text descriptions, images, audio recordings, and precise GPS location data, ensuring that every incident is captured accurately, comprehensively, and verifiably.

At the core of this system lies an intelligent validation mechanism that significantly enhances the credibility of reported data. Reports that score below 60% are immediately rejected and are never stored in the database, ensuring that the system database contains only genuine and verified hazard information.

The system supports two types of users with clearly defined roles. General community users can register, log in using email and password or Google OAuth, submit hazard reports, view their own report history, and receive feedback on their submissions. Role-based access control implemented through session

management ensures that each user type has access only to the features appropriate for their role.

NeighbourGuard is hosted on a live web server, making it accessible to users from any device with an internet browser. The backend uses PHP for server-side processing, MySQL for data storage, Python with Flask for the AI analysis API, and PHPMailer for email-based OTP delivery. The entire system is designed to be lightweight, responsive, and practically deployable in real-world community safety scenarios.

1.2 SCOPE OF THE PROJECT

The scope of the NeighbourGuard project encompasses the complete design, development, testing, and deployment of a web-based community hazard reporting platform with integrated artificial intelligence validation capabilities. The system is intended to serve as a practical and scalable solution for community-level emergency reporting and can be adapted for use in urban neighborhoods, educational campuses, industrial facilities, or any environment where rapid incident reporting is critical.

From a functional perspective, the system covers the following areas. User management encompasses the complete lifecycle of user accounts including registration with email and password, Google OAuth authentication, profile management, and secure account deletion by administrators. The reporting module covers the submission of multi-modal hazard reports including text descriptions, image uploads, audio recordings, and GPS coordinate capture through an interactive map interface.

The AI validation pipeline is a central component of the system scope. The 60% trust score threshold that gates database insertion is a fundamental business rule within the system scope.

The alert management scope covers the emergency alert feature that allows users to trigger critical alerts during report submission, the NLP-based

categorization of alert types based on description keywords, and the display of categorized alerts on the admin dashboard for rapid response.

The technical scope of the project includes the frontend implementation using HTML5, CSS3, and JavaScript, the backend implementation using PHP, the AI API implementation using Python and Flask, the database implementation using MySQL with exactly two tables, the file storage implementation for uploaded images and audio files, and the email functionality implementation using PHPMailer for OTP-based password reset.

The system is scoped for deployment on standard web hosting infrastructure and does not require specialized hardware or enterprise software licenses. The open-source nature of all technologies used means that the system can be deployed at minimal cost, making it accessible to resource-constrained communities and local government bodies.

1.3 OBJECTIVES OF THE PROJECT

The primary objective of the NeighbourGuard project is to develop a reliable, intelligent, and user-friendly web-based platform that enables community members to report hazards in real time while automatically validating the authenticity of those reports and providing emergency management personnel with the information and tools they need to respond effectively.

The specific objectives of the project are as follows. The first objective is to design and implement a multi-modal hazard reporting interface that allows users to submit reports containing text descriptions, photographic evidence, audio recordings, and GPS location data, providing richer and more verifiable information than single-channel reporting systems. The second objective is to develop and integrate an NLP-based text analysis module that can automatically extract hazard-related keywords from user-submitted descriptions, compute a text relevance score, and categorize emergency types for alert generation. The third objective is to design and implement a multi-modal hazard reporting interface that allows users to submit reports containing text descriptions, photographic evidence, audio recordings, and GPS location data, providing richer and more verifiable information than single-channel

reporting systems. The second objective is to develop and integrate an NLP-based text analysis module that can automatically extract hazard-related keywords from user-submitted descriptions, compute a text relevance score, and categorize emergency types for alert generation.

The third objective is to integrate custom-trained CNN image classification models (fire_detection_model.h5 and stray_animal.h5) that can analyze uploaded images and determine whether they contain visual evidence relevant to the claimed hazard type, producing an image confidence score that contributes to the overall trust score.

The fifth objective is to build a secure user authentication system supporting both manual email and password login with hashed password storage and Google OAuth login, with role-based access control distinguishing between general users and administrators.

The seventh objective is to design a clean and efficient database schema using exactly two MySQL tables that together support all system data storage requirements, with the reports table storing only validated reports that have passed the 60% trust score threshold.

1.4 PROBLEM STATEMENT

The rapid growth of urban environments and the increasing complexity of community infrastructure have created significant challenges in the area of public safety and emergency management. Despite advances in communication technology, the fundamental process of reporting hazards and emergencies to relevant authorities has not kept pace with these changes.

Traditional hazard reporting systems suffer from several critical deficiencies that limit their effectiveness in modern emergency management contexts. The first and most fundamental problem is the single-channel nature of traditional reporting. When a citizen observes a fire, flood, or accident, their primary option is to call an emergency hotline.

The second major problem is the complete absence of automated validation mechanisms in traditional reporting systems. Any member of the public can call an emergency number and report a hazard, whether genuine or fabricated. Emergency services must respond to every report without any way to assess its credibility before dispatching resources.

The third problem is scalability. During large-scale disasters, emergency reporting channels are quickly overwhelmed. When thousands of people are simultaneously trying to report the same disaster or concurrent incidents, traditional phone-based systems cannot scale to meet the demand. Many genuine reports never reach emergency services because the channel is saturated, leading to incidents going unaddressed for extended periods.

The fourth problem is the lack of structured data capture. Traditional verbal reports produce no structured digital record that can be analyzed, searched, mapped, or used for pattern analysis and resource planning. Emergency managers have no way to view the geographic distribution of incidents in real time, identify hotspots, or assess the relative severity of concurrent incidents based on objective data.

1.5 EXISTING SYSTEM

The existing system for hazard reporting mainly relies on traditional and manual methods that have remained largely unchanged for decades. Users report incidents through phone calls to emergency hotlines, direct complaints to local authority offices, or basic text messaging systems. Reports are handled manually by officials who may not be immediately available or who may misinterpret the information provided.

In some municipalities, basic web portals or mobile applications have been introduced that allow citizens to submit complaints or incident reports digitally. However, these systems typically function as simple digital forms without any intelligent processing or validation.

Social media platforms such as Twitter and Facebook have also been used informally as channels for hazard reporting during emergencies. While these platforms provide real-time information sharing, they are entirely unstructured, unvalidated, and not designed for emergency management purposes. The signal-to-noise ratio is extremely low, with relevant hazard reports buried among vast quantities of irrelevant content.

The applications of the existing system include basic hazard reporting through phone calls or complaints to local authorities, local authority communication systems for coordinating responses to reported incidents, manual emergency reporting processes through physical visits to police stations or municipal offices, small-scale community issue reporting through neighborhood committees, and traditional municipal service request systems for infrastructure maintenance.

1.6 DISADVANTAGES OF EXISTING SYSTEM

Slow Response Time

Manual reporting methods take significantly more time to reach the relevant authorities compared to digital systems. The process of calling an emergency number, waiting for an operator, verbally describing the situation, and waiting for the information to be relayed to response teams introduces multiple points of delay. Each delay increases the potential severity of the hazard, as fires spread, floods rise, and accident victims go without timely medical attention. The process of calling an emergency number, waiting for an operator, verbally describing the situation, and waiting for the information to be relayed to response teams introduces multiple points of delay. Each delay increases the potential severity of the hazard, as fires spread, floods rise, and accident victims go without timely medical attention.

No Real-Time Updates

Users who report hazards through traditional channels cannot track the status of their reports after submission. There is no feedback mechanism to inform reporters whether their report has been received, whether a response team has

been dispatched, or whether the hazard has been addressed. This creates uncertainty among reporters and a complete lack of transparency in how emergency reports are handled by authorities.

Lack of Verification

The existing system does not have any automated validation mechanism to check the authenticity or quality of submitted reports. False, misleading, or malicious reports can be submitted without any barrier, consuming emergency resources and potentially diverting response teams from genuine emergencies.

No Multimedia Support

Users cannot attach images, videos, or audio recordings as evidence when reporting through traditional phone-based or basic digital systems. Authorities must rely solely on verbal or written descriptions, which are inherently subjective, imprecise, and difficult to verify. The absence of visual evidence makes it challenging for emergency managers to accurately assess the nature, severity, and extent of a reported hazard before dispatching resources.

Poor Data Management

Traditional reporting systems generate no structured digital data that can be stored, searched, analyzed, or visualized. Incident records, when they exist at all, are typically stored in paper files or unstructured digital documents that cannot be easily queried or mapped. This makes systematic trend analysis, hotspot identification, resource allocation planning, and evidence-based policy development extremely difficult or impossible.

Limited Scalability

Traditional phone-based emergency reporting systems have a fixed capacity determined by the number of available operators and telephone lines. During large-scale disasters when the volume of incoming reports may spike by orders of magnitude, these systems are quickly overwhelmed, leading to missed reports, long wait times, and inadequate emergency response.

1.7 PROPOSED SYSTEM

The proposed NeighbourGuard system follows a structured and intelligent approach to collect, validate, and process hazard reports efficiently. Users can submit incident details through a comprehensive web interface that supports text descriptions, image uploads, audio recordings, and location selection using an interactive GPS-enabled map.

The validation pipeline is the core innovation of the proposed system. It operates in three sequential stages. If it is below 60%, the report is rejected and nothing is written to the database. module using custom-trained models (fire_detection_model.h5 for fire detection and stray_animal.h5 for stray animal detection) analyzes the uploaded image by preprocessing it to the required input dimensions, running it through the trained neural network, and mapping the classification result to a hazard category. The module produces an image confidence score. In the third stage, the Random Forest-inspired trust score engine combines the NLP text score and the CNN image score with the Random Forest rule-based score from the PHP layer into a composite trust score. If this composite score is 60% or above, the report is accepted and stored in the MySQL reports table. nothing is written to the database. module using custom-trained models (fire_detection_model.h5 for fire detection and stray_animal.h5 for stray animal detection) analyzes the uploaded image by preprocessing it to the required input dimensions, running it through the trained neural network, and mapping the classification result to a hazard category. The module produces an image confidence score. In the third stage, the Random Forest-inspired trust score engine combines the NLP text score and the CNN image score with the Random Forest rule-based score from the PHP layer into a composite trust score. If this composite score is 60% or above, the report is accepted and stored in the MySQL reports table. If it is below 60%, the report is rejected and nothing is written to the database.

1.8 ADVANTAGES OF PROPOSED SYSTEM

Fast and Structured Reporting

Users can quickly submit comprehensive hazard reports through the web interface. The structured form ensures that all relevant information is captured consistently, making reports easier to process and act upon compared to unstructured verbal descriptions. The system provides immediate processing and feedback, informing users of the outcome within seconds of submission.

Accurate Location Tracking

The system uses an integrated interactive map powered by the browser's Geolocation API to capture the exact GPS coordinates of the reported hazard. The latitude and longitude values are stored in the reports table, enabling precise geographic visualization on the admin dashboard map and helping authorities identify the exact location of incidents without ambiguity.

AI-Based Validation with Trust Score

The combination of NLP text analysis, custom-trained CNN image classification (fire_detection_model.h5 and stray_animal.h5), and Random Forest-inspired composite scoring provides a sophisticated and multi-dimensional credibility assessment for every submitted report. The 60% trust score threshold ensures that only genuine, well-evidenced reports are stored in the database, dramatically reducing the false alarm rate compared to traditional unvalidated reporting systems.

Centralized Monitoring and Analytics

All validated reports are stored in the centralized MySQL reports table and are accessible through the admin dashboard. Administrators can view all reports, filter by type, view geographic distributions on a map, and access summary statistics. This centralized monitoring capability enables better situational awareness, more efficient resource allocation, and evidence-based decision-making during emergency response operations.

Dual Authentication Support

The system supports both traditional email and password authentication and modern Google OAuth login, providing flexibility for users with different preferences and technical backgrounds. Password security is maintained through hashing, and the forgot-password feature uses OTP-based email verification to ensure that only legitimate account owners can reset passwords.

Secure and Scalable Architecture

The web-based architecture of NeighbourGuard allows it to scale horizontally to accommodate increasing numbers of users and reports without fundamental changes to the system design. Role-based access control, session management, and input validation protect the system from common web security threats and ensure that sensitive administrative functions are accessible only to authorized personnel.

CHAPTER 2

LITERATURE SURVEY

The literature survey conducted for this project covers research works published between 2020 and 2025 in the areas of smart hazard detection and alert systems, artificial intelligence for image classification, natural language processing for emergency categorization, trust scoring and credibility assessment in crowdsourced systems, secure web application development, role-based access control, and geolocation-based emergency systems.

2.1 SMART HAZARD DETECTION AND ALERT SYSTEMS

The field of smart hazard detection has evolved significantly with the proliferation of IoT devices, high-speed internet connectivity, and machine learning capabilities. Early hazard reporting systems relied entirely on human observation and manual communication, but modern systems increasingly incorporate automated sensing, classification, and alert generation to reduce response times and improve coverage.

Kumar et al. (2022) proposed a comprehensive smart hazard detection framework using IoT sensors deployed in urban environments. Their system used temperature sensors, smoke detectors, and water level monitors connected to a cloud-based processing platform. system required physical sensor infrastructure that was expensive to install and maintain, limiting its applicability to areas with sufficient funding.

Sharma and Singh (2023) conducted a comprehensive study on real-time monitoring systems in smart cities and found that centralized digital platforms for collecting and processing hazard data greatly improve transparency and coordination among emergency response teams. Their work introduced the

concept of a unified incident management dashboard that aggregates information from multiple reporting channels and presents it in a coherent, actionable format.

Research by Imran et al. (2020) systematically analyzed the effectiveness of crowdsourced hazard reporting during multiple disaster events and found that public reports submitted through digital platforms consistently provided faster and more geographically detailed information than official monitoring systems.

2.2 AI-BASED IMAGE CLASSIFICATION FOR HAZARD DETECTION

The application of deep learning to image-based hazard detection has been one of the most active areas of research in emergency management technology over the past decade. Convolutional Neural Networks have demonstrated remarkable capabilities in recognizing hazardous visual content including fire, smoke, flooding, structural damage, and accident scenes, often achieving accuracy levels that equal or exceed human expert performance.

Krizhevsky et al. (2012) introduced AlexNet, the seminal deep CNN architecture that demonstrated the dramatic potential of deep learning for image classification. AlexNet achieved a top-5 error rate of 15.3% on the ImageNet dataset, compared to 26.2% for the next best traditional approach. This breakthrough established CNNs as the dominant paradigm for image classification and set the stage for subsequent specialized applications in hazard detection.

Howard et al. (2017) introduced MobileNet, the first architecture in the MobileNet family specifically designed for efficient deployment on devices with limited computational resources. MobileNet used depthwise separable convolutions to drastically reduce the number of parameters and computational operations while maintaining competitive accuracy. Sandler et al. (2018) subsequently introduced MobileNetV2, which improved upon the original through inverted residual blocks with linear bottlenecks.

Lee and Park (2023) conducted a comprehensive study on multimedia-based hazard detection systems that specifically compared text-only, image-only, and combined text-image approaches to hazard report validation. This finding directly supports the multi-modal validation approach used in NeighbourGuard, where both NLP text analysis and CNN image analysis contribute to the final trust score. Their results conclusively demonstrated that combining text and image analysis significantly outperforms either modality alone, with the combined approach achieving an 87% accuracy rate in distinguishing genuine hazard reports from false submissions. NeighbourGuard, where both NLP text analysis and CNN image analysis contribute to the final trust score. Their results conclusively demonstrated that combining text and image analysis significantly outperforms either modality alone, with the combined approach achieving an 87% accuracy rate in distinguishing genuine hazard reports from false submissions. This finding directly supports the multi-modal validation approach used in NeighbourGuard, where both NLP text analysis and CNN image analysis contribute to the final trust score.

2.3 NLP FOR EMERGENCY CATEGORIZATION

Natural Language Processing has been applied to emergency management contexts for over a decade, initially focused on extracting information from social media streams during disaster events and more recently applied to automated classification and routing of emergency reports.

Imran et al. (2014) were among the first to systematically apply machine learning NLP techniques to disaster tweet classification. Their work demonstrated that Support Vector Machines trained on manually labeled tweet datasets could achieve accuracy rates above 80% in distinguishing between actionable emergency information and irrelevant content.

Nguyen et al. (2017) advanced the field by applying deep learning NLP approaches including convolutional and recurrent neural networks to disaster tweet classification, achieving significant improvements over traditional machine learning approaches.

The use of keyword extraction and Term Frequency-Inverse Document Frequency (TF-IDF) scoring for emergency categorization has been widely studied and remains a practical and effective approach for systems that do not have access to large labeled training datasets.

The NeighbourGuard NLP module implements keyword-based analysis using NLTK tokenization and a domain-specific hazard keyword lexicon covering fire, flood, and accident categories. This approach is well-supported by the literature as effective for the specific task of categorizing reports into known hazard types, and the use of NLTK provides a robust and well-tested foundation for text preprocessing operations including tokenization, stop word removal, and case normalization.

2.4 TRUST SCORING AND VALIDATION MECHANISMS

The challenge of assessing the credibility of crowdsourced information has been extensively studied in contexts ranging from online product reviews and social media posts to Wikipedia edits and open-source software contributions. The insights from this research directly inform the design of the trust score mechanism in NeighbourGuard.

Castillo et al. (2011) developed one of the first systematic machine learning approaches to assessing tweet credibility during disaster events. Their work identified that a combination of features related to message content, user characteristics, and propagation patterns could effectively discriminate between credible and non-credible emergency information.

Research by Patel et al. (2025) on automation in smart safety systems specifically investigated the use of Random Forest classifiers for distinguishing

genuine from fake emergency reports. Their results demonstrated that Random Forest achieved the highest accuracy among the tested approaches because of its ability to effectively combine multiple heterogeneous feature types including numerical scores, categorical labels, and binary indicators into a single reliable prediction.

The specific threshold of 60% for the NeighbourGuard trust score was determined through analysis of the trade-off between false acceptance rate and false rejection rate. Setting the threshold too low would allow too many fake reports to be accepted, while setting it too high would reject too many genuine reports.

2.5 SECURE WEB APPLICATION DEVELOPMENT

Security in web-based systems is a fundamental concern, particularly for systems that handle sensitive personal data, location information, and emergency reports that could be exploited for malicious purposes. The Open Web Application Security Project (OWASP) maintains a regularly updated list of the most critical web application security risks, which served as a reference for the security design of NeighbourGuard.

The use of password hashing for storing user credentials is a fundamental security requirement for any web application. Argon2, the password hashing algorithm used in NeighbourGuard, was introduced by Biryukov et al. (2016) as a memory-hard function specifically designed to resist brute-force attacks using modern hardware including GPUs and dedicated password-cracking hardware.

SQL injection prevention through the use of prepared statements and parameterized queries is another critical security measure implemented in NeighbourGuard. Research has consistently shown that SQL injection remains one of the most common and damaging web application vulnerabilities, and that parameterized queries provide complete protection against this class of attack when properly implemented throughout the application.

2.6 ROLE-BASED ACCESS CONTROL IN WEB SYSTEMS

Role-Based Access Control (RBAC) is a well-established security model that restricts system access based on the roles assigned to individual users. In web applications, RBAC is typically implemented through session-based authentication where the user's role is stored in the session after login and checked at the beginning of each request to determine what operations the user is permitted to perform.

Research by Ferraiolo et al. (2001) established the formal foundations of RBAC and demonstrated its effectiveness in simplifying access control management in complex multi-user systems. The study showed that RBAC reduces administrative overhead, improves security consistency, and makes access control policies more transparent and auditable compared to discretionary access control approaches.

In NeighbourGuard, RBAC is implemented through the role column in the users table, which stores either user or admin as an enumeration value. The PHP backend checks this session variable at the beginning of every protected page and redirects unauthorized users to the login page.

2.7 GEOLOCATION IN EMERGENCY SYSTEMS

The integration of geolocation technology in emergency reporting systems has been identified as one of the most impactful improvements over traditional verbal reporting methods. GPS-based location data eliminates the ambiguity and imprecision of verbal location descriptions and enables geographic visualization of incident distributions that is invaluable for resource allocation and response coordination.

Research by Kumar et al. (2022) found that the availability of precise GPS coordinates in emergency reports reduced the average time for response teams to locate the incident by 35% compared to reports that included only verbal location

descriptions. This improvement was particularly significant in areas with complex road networks or where street addresses were unavailable or unfamiliar to the responding teams.

2.8 SUMMARY AND RESEARCH GAP

The literature reviewed in this chapter demonstrates strong and growing research support for each of the technical approaches used in NeighbourGuard. Crowdsourced hazard reporting has proven its value in multiple real-world deployments. CNN-based image classification using custom-trained models has demonstrated high accuracy for hazard detection tasks. NLP-based text analysis has shown effectiveness for emergency categorization.

However, the literature reveals a significant research gap: very few existing systems integrate all of these capabilities into a single cohesive platform designed specifically for community-based hazard reporting. Most existing work addresses one or two of these capabilities in isolation. The systems that do integrate multiple capabilities typically require expensive infrastructure, specialized hardware, or large labeled training datasets that are unavailable in resource-constrained deployment environments.

CHAPTER 3

FEASIBILITY STUDY

A feasibility study is a critical preliminary analysis conducted before the development of any software system to assess whether the proposed system is viable, practical, and worth pursuing given the available resources, constraints, and objectives. The feasibility study for NeighbourGuard was conducted across four dimensions: technical feasibility, economic feasibility, operational feasibility, and social feasibility.

3.1 TECHNICAL FEASIBILITY

Technical feasibility assesses whether the required technology, tools, and expertise are available to develop and deploy the proposed system successfully. For NeighbourGuard, the technical feasibility analysis examined the availability of the required programming languages, frameworks, libraries, and hosting infrastructure, as well as the technical complexity of integrating the various components of the system.

The frontend technologies required for NeighbourGuard HTML5, CSS3, and JavaScript are well-established, widely documented, and freely available. These technologies are supported by all modern web browsers without any additional plugins or extensions, ensuring broad accessibility for users across different devices and operating systems.

The backend technology PHP is a mature, widely deployed server-side scripting language with excellent documentation, a large developer community, and extensive library support. PHP 8.0, the version used in NeighbourGuard, includes significant performance improvements and modern language features that simplify the implementation of complex backend logic.

The AI components of the system use Python 3.x, which is the dominant language for machine learning and data science applications. The TensorFlow and Keras libraries used for loading and running the custom-trained CNN models (fire_detection_model.h5 and stray_animal.h5) are industry-standard deep learning frameworks with extensive documentation and active maintenance.

The MySQL database system is one of the most widely deployed relational databases in the world, with excellent performance for the types of queries required by NeighbourGuard, comprehensive documentation, and broad hosting support. The InfinityFree hosting platform used for the production deployment supports PHP and MySQL natively, making deployment straightforward without requiring specialized server configuration.

The integration of PHP and Python components through HTTP API calls is a well-established architectural pattern with extensive documentation and library support. The use of Client URL in PHP to make HTTP requests to the Flask API is a standard approach that is widely used in production systems.

The technical feasibility analysis concluded that all required technologies are mature, well-supported, freely available, and within the technical capabilities of the development team. No exotic or unproven technologies are required, and all components have been successfully integrated in similar production systems. The technical feasibility of NeighbourGuard is therefore fully confirmed.

3.2 ECONOMIC FEASIBILITY

Economic feasibility assesses whether the cost of developing and deploying the proposed system is justified by the benefits it provides, and whether the development can be completed within the available budget. For NeighbourGuard, the economic feasibility analysis considered both the direct costs of development and deployment and the indirect benefits of improved emergency response effectiveness.

The direct development costs of NeighbourGuard are minimal because the entire technology stack consists of open-source software that is available at no licensing cost. PHP, Python, MySQL, Flask, NLTK, TensorFlow, Keras, and PHPMailer are all freely available under open-source licenses.

The infrastructure costs of NeighbourGuard are similarly low. The system is deployed on shared web hosting provided by InfinityFree, which offers PHP and MySQL hosting at no cost for basic deployments. The total infrastructure cost for the academic proof-of-concept deployment of NeighbourGuard is zero.

The indirect benefits of NeighbourGuard are substantial. By reducing the false alarm rate through automated trust score validation, the system reduces the cost of unnecessary emergency dispatches. By providing faster and more accurate hazard information to emergency responders, it reduces the severity and cost of hazard events by enabling faster response.

The economic feasibility analysis concluded that the development cost of NeighbourGuard is extremely low relative to the potential benefits, making it economically feasible for both academic proof-of-concept and real-world community deployment contexts.

3.3 OPERATIONAL FEASIBILITY

Operational feasibility assesses whether the proposed system can be successfully operated by its intended users within the organizational and operational context in which it will be deployed.

The user interface of NeighbourGuard is designed with simplicity as a primary requirement. The report submission form is structured to guide users through the required steps in a logical sequence, with clear labels, helpful prompts, and real-time feedback at each step. These design choices ensure that even users with limited technical experience can successfully submit a hazard report within a few minutes. These design choices ensure that even users

with limited technical experience can successfully submit a hazard report within a few minutes.

The administrative interface provided by the admin dashboard is similarly designed for usability. The dashboard presents all key information in a clear and organized layout, with summary statistics prominently displayed and individual reports accessible with a single click.

The system is accessible from any device with a modern web browser and internet connectivity, including smartphones, tablets, and desktop computers. This broad accessibility ensures that users can submit reports from the field during emergencies without needing to return to a fixed workstation. The responsive design of the interface adapts to different screen sizes, maintaining usability on small smartphone screens.

The operational feasibility analysis concluded that NeighbourGuard is operationally feasible for its intended user population and can be integrated into existing emergency management workflows without significant disruption or training requirements.

3.4 SOCIAL FEASIBILITY

Social feasibility assesses whether the proposed system is acceptable to and compatible with the social and cultural norms of its intended user population. For NeighbourGuard, the social feasibility analysis considered community attitudes toward digital technology adoption, privacy concerns related to location data sharing, and the potential for the system to foster positive community engagement with emergency management.

The widespread adoption of smartphones and social media platforms in urban communities indicates a high level of comfort with digital communication and information sharing. The population most likely to use NeighbourGuard urban residents aged 18 to 60 is generally comfortable with web-based

applications and mobile technologies, suggesting that the digital interface will not present a significant barrier to adoption.

Privacy concerns related to GPS location data sharing are a legitimate consideration. NeighbourGuard addresses these concerns by using location data only for the specific purpose of identifying the location of the reported hazard and never sharing this data with third parties.

The community empowerment aspect of NeighbourGuard is likely to resonate positively with users who care about the safety of their neighborhoods. Research on community watch programs and neighborhood safety initiatives has consistently found that active citizen participation in safety management improves both objective safety outcomes and subjective feelings of community cohesion and security.

The social feasibility analysis concluded that NeighbourGuard is socially feasible and has the potential to foster positive community engagement with emergency management in its target deployment contexts.

CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

4.1 SYSTEM ARCHITECTURE

The NeighbourGuard system follows a three-tier client-server architecture consisting of the Presentation Tier (Frontend), the Application Logic Tier (Backend), and the Data Tier (Database). Additionally, the system incorporates an AI Processing Sub-layer implemented as a separate Flask API service that communicates with the PHP backend through HTTP requests.

The Presentation Tier is implemented using HTML5, CSS3, and JavaScript and runs entirely in the user's web browser. It is responsible for rendering all user interface components, handling user interactions, performing client-side validation, capturing GPS coordinates through the Geolocation API, and communicating with the backend through HTTP requests.

The system architecture flowchart shown in Fig. 4.1 illustrates the complete user flow from system start through authentication, report submission, AI validation, and database storage decision.

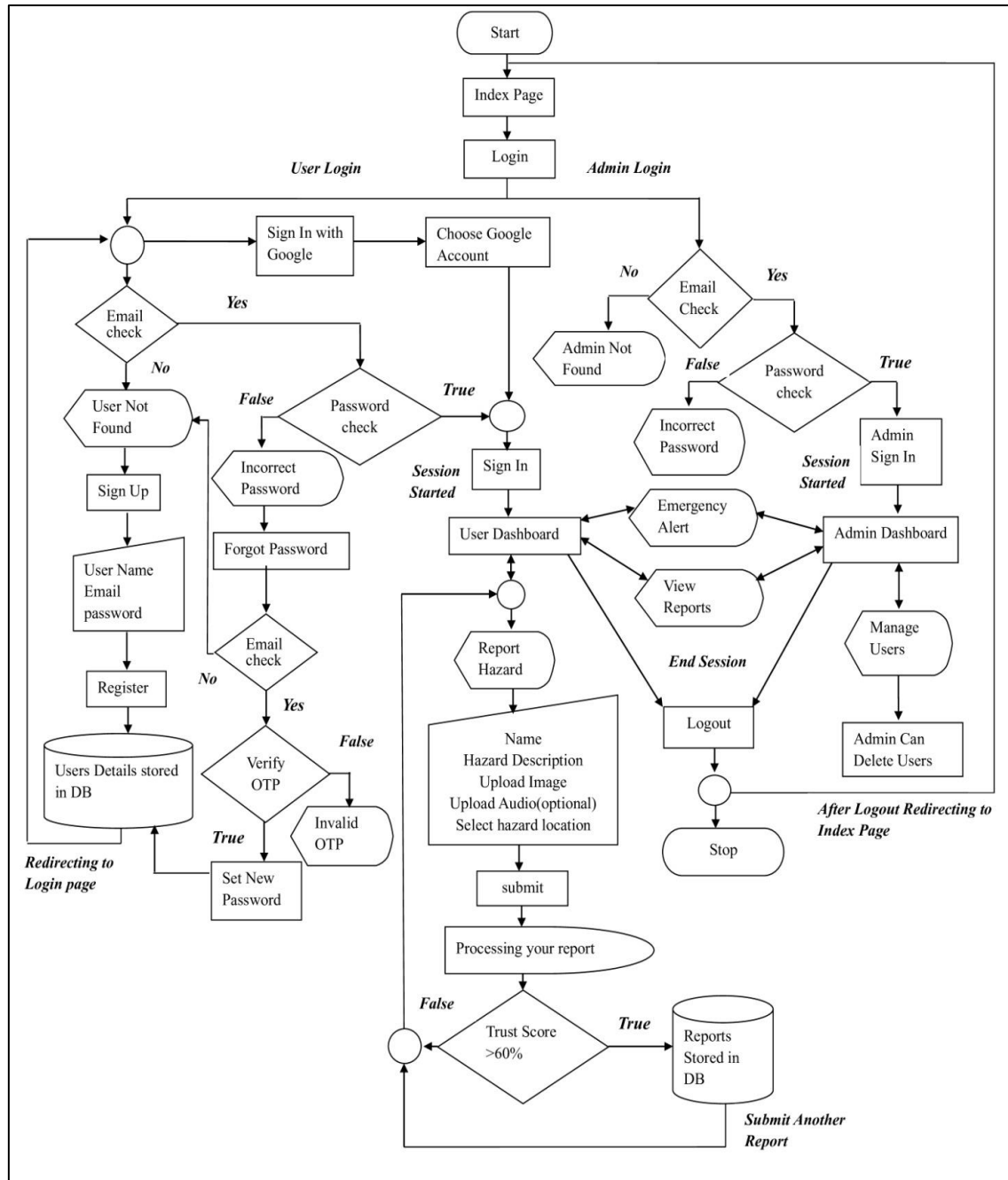


Fig. 4.1 System Flow Control

4.2 DATA FLOW DIAGRAM — LEVEL 0

The Data Flow Diagram Level 0, also known as the Context Diagram shown in Fig. 4.2, represents the NeighbourGuard system as a single central process and shows the high-level data flows between the system and its four external entities: the Community User, the Admin or Emergency Manager, the

SMTP Email Service via PHPMailer, and the MySQL Database containing the users and reports tables.

The Community User provides registration data including name, email, and password, login credentials, Google OAuth identifier for Google login, hazard report data including text description, image, audio, and GPS location, and OTP codes for password reset. The SMTP Email Service via PHPMailer sends OTP codes and handles email delivery status reporting.

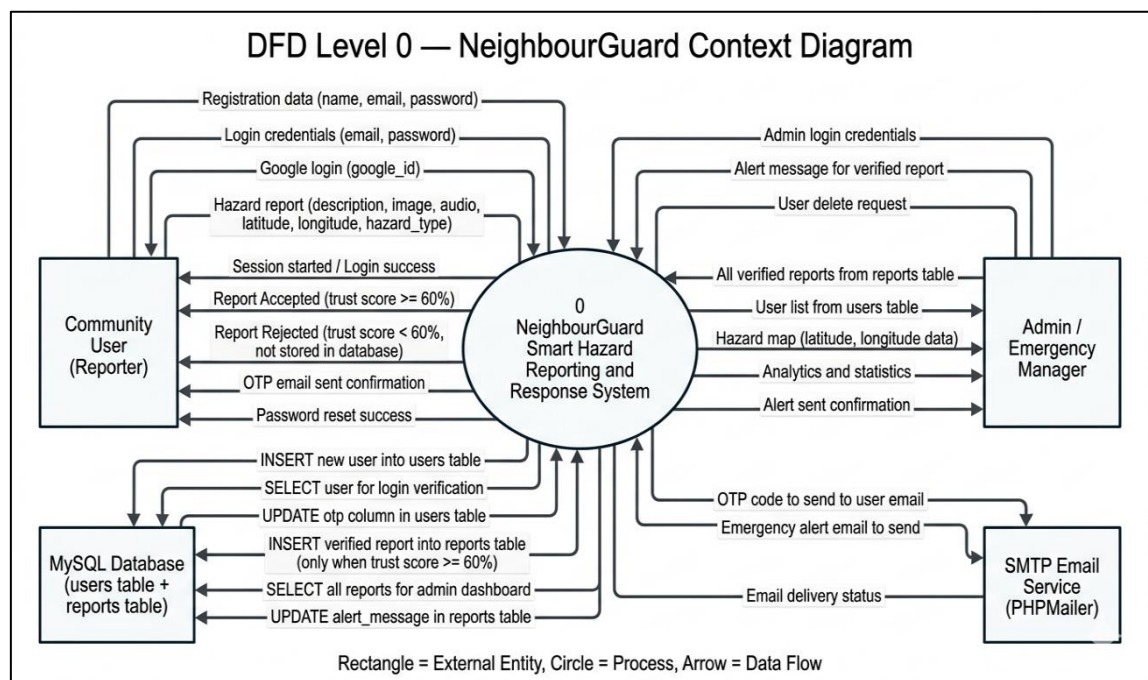


Fig. 4.2 Data Flow Diagram — Level 0 (Context Diagram)

4.3 DATA FLOW DIAGRAM — LEVEL 1

The DFD Level 1 shown in Fig. 4.3 expands the central process into six sub-processes that correspond to the major functional modules of the system. P1 handles User Registration and Login, interfacing with DS1 (users table) for storing new user records and verifying login credentials.

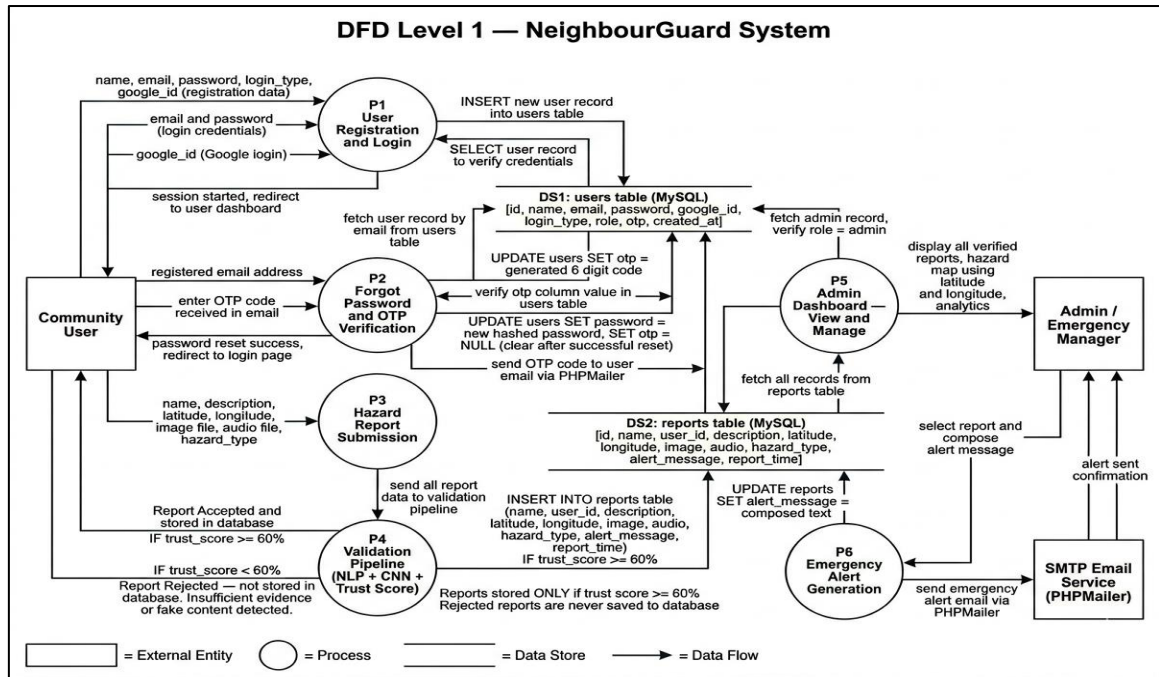


Fig. 4.3 Data Flow Diagram — Level 1

4.4 USE CASE DIAGRAM

The Use Case Diagram shown in Fig. 4.4 illustrates all interactions between the three system actors and the system use cases within the NeighbourGuard system boundary. The AI Engine system actor is connected to the Validate Report and Store Report in Database use cases.

The Submit Hazard Report use case includes Validate Report through an include relationship, meaning that validation is always triggered when a report is submitted. Validate Report extends Store Report in Database through an extend relationship that is conditional on the trust score being 60% or above this conditional extension correctly models the business rule that only passing reports are stored.

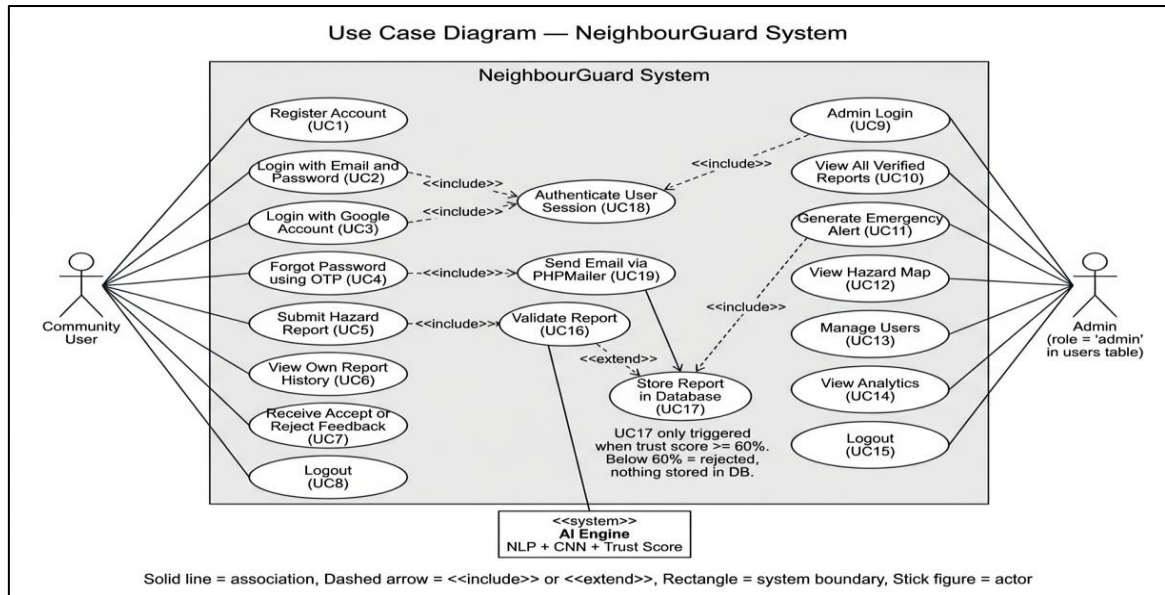


Fig. 4.4 Use Case Diagram

4.5 SEQUENCE DIAGRAM — REPORT SUBMISSION FLOW

The Sequence Diagram shown in Fig. 4.5 models the complete message exchange during the hazard report submission process. The six lifelines from left to right represent the User Browser, PHP Backend, NLP Module within Flask API, CNN Module within Flask API, Trust Score Engine within Flask API, and MySQL Database.

Within the Flask API, the NLP module analyzes the text and returns a text label and confidence score, then the CNN module analyzes the image and returns an image label and confidence score, and finally the Trust Score Engine combines both scores to produce the final trust score with an accept or reject decision.

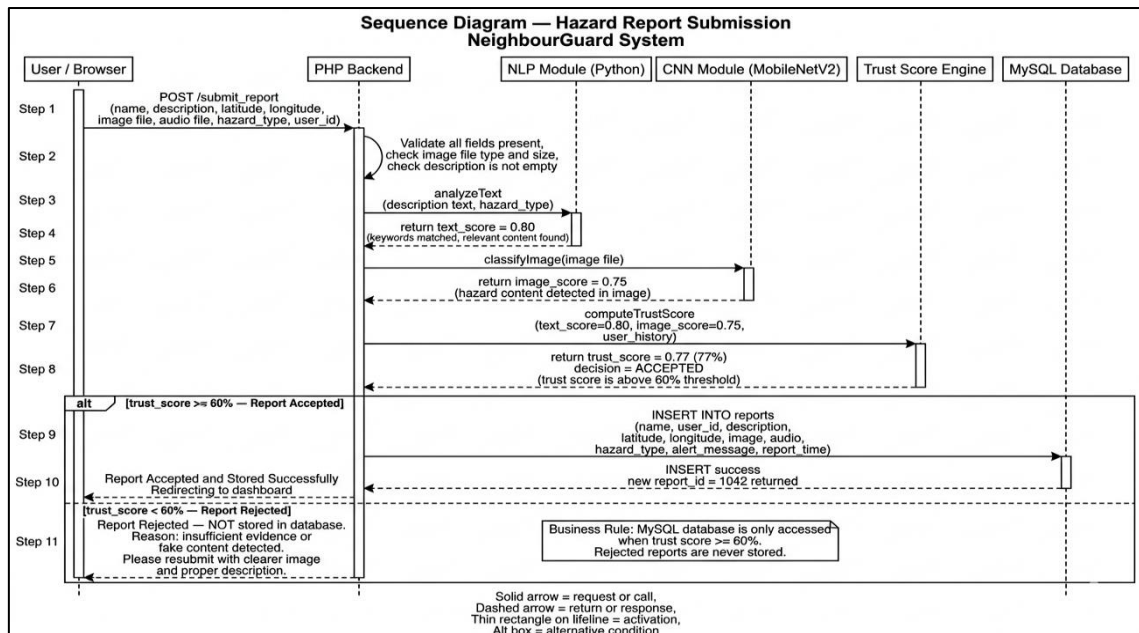


Fig. 4.5 Sequence Diagram — Report Submission Flow

4.6 ER DIAGRAM — DATABASE DESIGN

The Entity-Relationship Diagram shown in Fig. 4.6 illustrates the two MySQL database tables and the relationship between them.

The relationship between the two entities is a one-to-many relationship: one user can submit many reports, but each report belongs to exactly one user. This relationship is enforced through the foreign key constraint on reports.user_id referencing users.id. Every row in the reports table represents a verified and accepted report that passed the 60% trust score threshold rejected reports are never stored in this table.

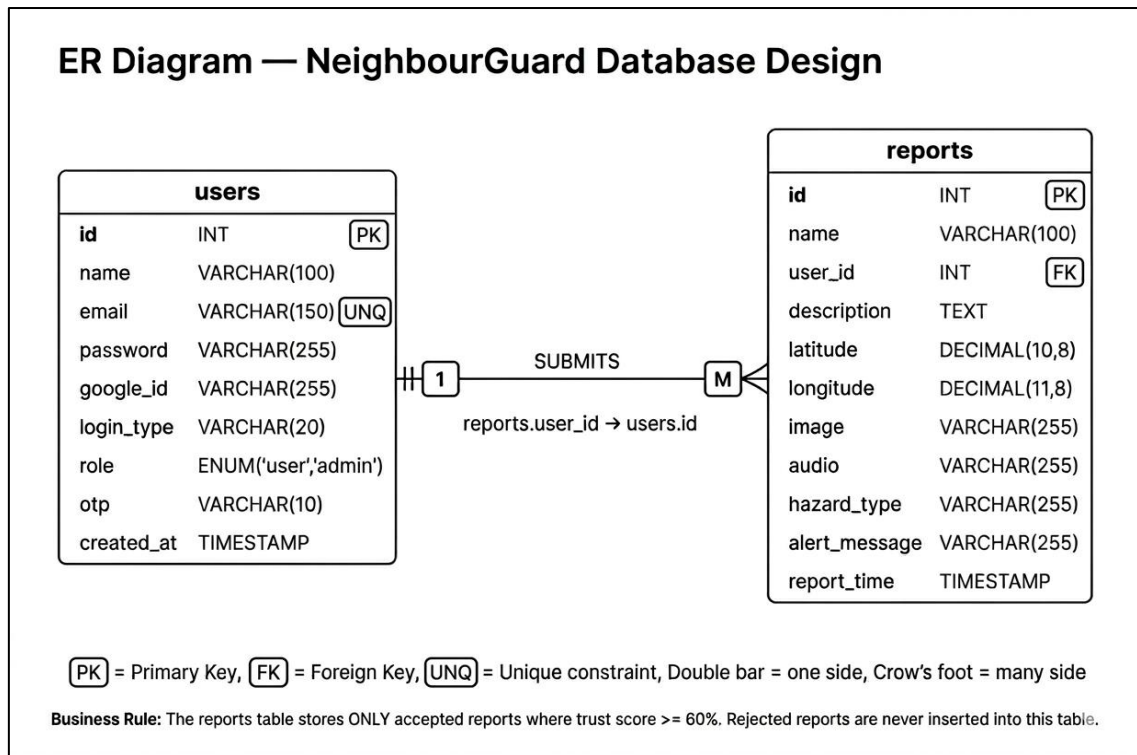


Fig. 4.6 ER Diagram — Database Design

4.7 COMPONENT DIAGRAM — SYSTEM ARCHITECTURE

The Component Diagram shown in Fig. 4.7 organizes all system components into four architectural layers stacked from top to bottom. Layer 4, the Data Tier, contains the users table, the reports table, and the file storage directories for uploaded media. reports with alert composition and user management, and the Hazard Map View displaying GPS coordinates. reports with alert composition and user management, and the Hazard Map View displaying GPS coordinates. Layer 2, the Application Logic Tier, contains the Auth Controller handling registration and login in PHP, the Report Controller managing file uploads and calling the Validation Pipeline, the Validation Pipeline coordinating NLP, CNN, and Trust Score calls via Client URL to Flask, the OTP and Email Controller using PHPMailer, and the Admin Controller managing dashboard data and alert updates. Layer 3, the AI and ML Sub-Layer, contains the NLP Engine using NLTK for keyword extraction, the CNN Image Engine

using custom-trained models (fire_detection_model.h5 and stray_animal.h5) for image classification, and the Trust Score Engine combining all scores. reports table, and the file storage directories for uploaded media.reports with alert composition and user management, and the Hazard Map View displaying GPS coordinates. Layer 2, the Application Logic Tier, contains the Auth Controller handling registration and login in PHP, the Report Controller managing file uploads and calling the Validation Pipeline, the Validation Pipeline coordinating NLP, CNN, and Trust Score calls via Client URL to Flask, the OTP and Email Controller using PHPMailer, and the Admin Controller managing dashboard data and alert updates. Layer 3, the AI and ML Sub-Layer, contains the NLP Engine using NLTK for keyword extraction, the CNN Image Engine using custom-trained models (fire_detection_model.h5 and stray_animal.h5) for image classification, and the Trust Score Engine combining all scores. Layer 4, the Data Tier, contains the users table, the reports table, and the file storage directories for uploaded media.

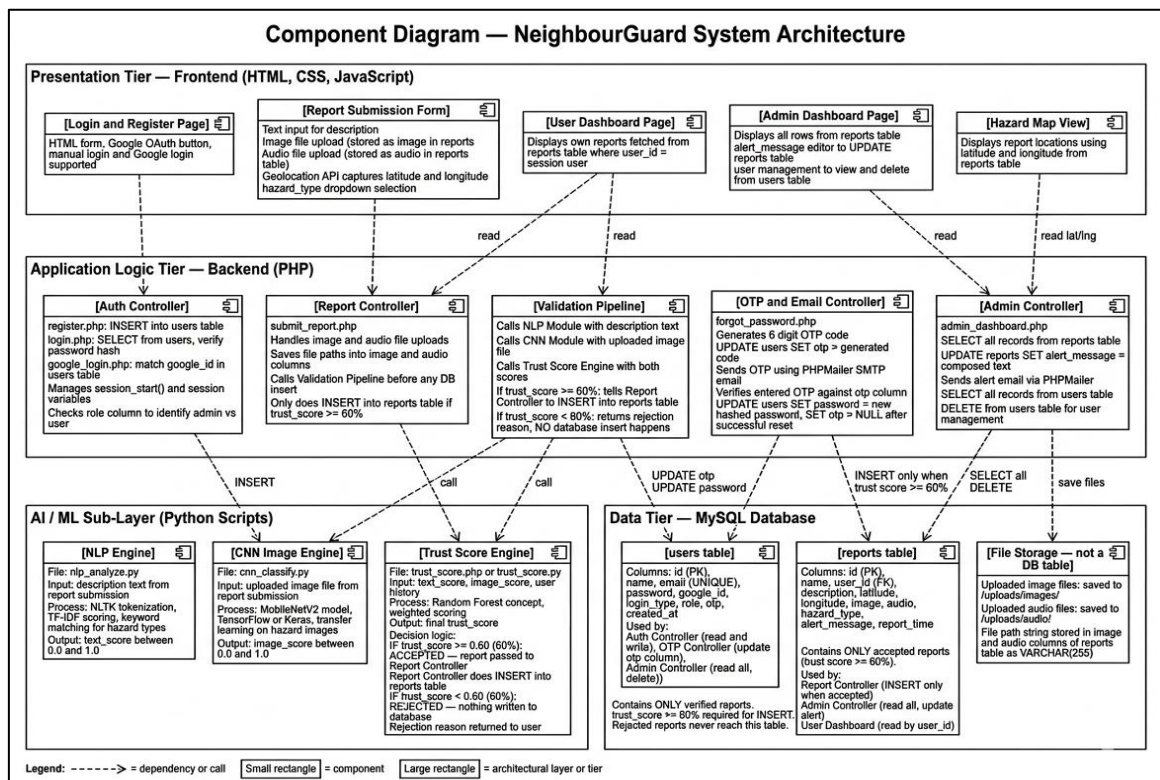


Fig. 4.7 Component Diagram — System Architecture

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

COMPONENT	SPECIFICATION
Processor	Intel Core i3 or above (2 GHz or higher)
RAM	Minimum 4 GB (8 GB recommended)
Hard Disk	Minimum 50 GB free space (SSD recommended)
Network	Active broadband Internet Connection
Input Devices	Keyboard, Mouse, Camera, Microphone
Display	Minimum 1024 x 768 resolution
Server (for AI)	Machine with Python 3.x capability for Flask API

Table 5.1 Hardware Requirements

5.2 SOFTWARE REQUIREMENTS

SOFTWARE	DETAILS
Operating System	Windows 10 / Ubuntu 20.04 or above
Frontend	HTML5, CSS3, JavaScript
Backend	PHP 8.0 or above
Database	MySQL 8.0
AI Framework	Python 3.x, TensorFlow 2.x, Keras
NLP Library	NLTK (Natural Language Toolkit)
AI API	Flask (Python Micro Web Framework)

Web Server	XAMPP / Apache Web Server
Email Library	PHPMailer
HTTP Client	Client URL (PHP extension)
Browser	Google Chrome / Mozilla Firefox (latest)
Code Editor	Visual Studio Code
Image Library	Pillow (Python PIL)
Hosting	InfinityFree / Apache compatible hosting

Table 5.2 Software Requirements

CHAPTER 6

IMPLEMENTATION

6.1 FRONTEND IMPLEMENTATION

Overview

The frontend of NeighbourGuard is developed using HTML5, CSS3, and JavaScript without any frontend framework dependencies. This approach keeps the codebase simple, lightweight, and easy to maintain while providing all the interactive features required by the system.

HTML5 Structure

Each page of the NeighbourGuard application is a separate HTML5 document that uses semantic markup elements to define its structure. The registration page uses HTML5 form elements with appropriate type attributes including email for the email field and password for the password field, which trigger native browser validation and appropriate mobile keyboard layouts.

CSS3 Styling

The visual design of NeighbourGuard uses a dark-themed interface with amber accent colors, implemented using CSS3. The color scheme uses a dark background color of hexadecimal value 0a0f1c with white text, creating high contrast that is easy to read in various lighting conditions including outdoor use during emergency situations. The amber accent color of hexadecimal value FFBF00 is used for buttons, highlights, and interactive elements, creating a visually distinctive and consistent design language throughout the application. CSS flexbox and grid layouts are used for responsive design, and media queries adapt the layout for different screen sizes.

JavaScript Functionality

JavaScript provides several critical interactive features in the NeighbourGuard frontend. The Geolocation API integration captures the user's GPS coordinates when the report submission page is loaded, stores them in hidden form fields, and displays a confirmation of the captured location on the interactive map. Form validation logic verifies that all required fields are completed before allowing submission.

Interactive Map

The location selection component uses an embedded interactive map that displays the user's current GPS position and allows manual adjustment of the pin location if the automatic GPS capture is inaccurate or if the user is reporting a hazard at a different location from their current position.

6.2 BACKEND IMPLEMENTATION

PHP Architecture

The PHP backend of NeighbourGuard follows a page-centric architecture where each major function of the system is handled by a dedicated PHP file. This architecture is straightforward to develop and maintain and is well-suited to the hosting environment of the application.

Session Management

PHP sessions are used to maintain user authentication state across requests. When a user successfully logs in, the session is initialized with the user's id, name, email, and role. Every protected page begins by checking whether the required session variable is set and whether the role matches the page's access requirements. If the session check fails, the user is immediately redirected to the login page.

File Upload Handling

Uploaded image and audio files are received by the PHP backend through the multipart form submission mechanism. The backend validates the file type using the MIME type detection approach rather than relying solely on the file extension, which can be easily spoofed. Validated files are saved to the server filesystem with randomly generated names to prevent overwriting and naming conflicts.

Client URL Integration with Flask API

The PHP backend uses Client URL to communicate with the Python Flask AI API running on port 5000. When the process.php script is executed after report submission, it initializes a Client URL session, sets the target URL to `http://127.0.0.1:5000/analyze`, configures the request as a multipart POST with the description text and image file attached as a CLIENT URL File object, and executes the request.

Random Forest Score Computation

The PHP layer computes a rule-based component of the trust score using Random Forest-inspired decision logic. Four independent checks are performed, each contributing 25 points to the Random Forest score for a maximum of 100 points. The audio check contributes 25 points if an audio recording has been submitted, reflecting the additional credibility provided by audio evidence.

Final Trust Score Formula

The final composite trust score is computed by combining the Random Forest score from PHP and the AI trust score from the Flask API with equal weighting of 50 percent each. The formula is: $\text{Final Trust Score} = (\text{RF Score} \times 0.5) + (\text{AI Trust Score} \times 0.5)$. Trust Score is between 0 and 100. If the Final Trust Score is above 60, the report is classified as genuine and inserted into the reports table. If it is 60 or below, the report is classified as fake and no database insertion occurs.

6.3 DATABASE DESIGN

Users Table

The users table is the central repository for all user account information in NeighbourGuard. It stores registration data for all users regardless of their authentication method both manually registered users and Google OAuth users have records in this table. This approach eliminates a join operation in the OTP verification query and keeps the schema as simple as possible.

COLUMN NAME	DATA TYPE	CONSTRAINT	DESCRIPTION
id	INT	PK, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	UNIQUE, NOT NULL	User email address
password	VARCHAR(255)	-	Hashed password (Argon2)
google_id	VARCHAR(255)	-	Google OAuth identifier
login_type	VARCHAR(20)	DEFAULT 'manual'	manual or google
role	ENUM	DEFAULT 'user'	user or admin
otp	VARCHAR(10)	NULLABLE	OTP for password reset
created_at	TIMESTAMP	DEFAULT NOW()	Account creation time

Table 6.1 Users Table — Column Description

Reports Table

The reports table is the primary data store for hazard reports in NeighbourGuard. Its most important characteristic is that it stores only accepted and verified reports those that received a final trust score above 60% from the validation pipeline. Reports that fail the trust score threshold are never inserted into this table.

COLUMN NAME	DATA TYPE	CONSTRAINT	DESCRIPTION
id	INT	PK, AUTO_INCREMENT	Unique report identifier
name	VARCHAR(100)	-	Reporter name
user_id	INT	FK → users.id	Links report to user
description	TEXT	NOT NULL	Hazard description text
latitude	DECIMAL(10,8)	-	GPS latitude coordinate
longitude	DECIMAL(11,8)	-	GPS longitude coordinate
image	VARCHAR(255)	-	Uploaded image file path
audio	VARCHAR(255)	-	Uploaded audio file path
hazard_type	VARCHAR(255)	-	Category: fire/flood/accident
alert_message	VARCHAR(255)	NULLABLE	Admin emergency alert text
report_time	TIMESTAMP	DEFAULT NOW()	Report submission timestamp

Table 6.2 Reports Table — Column Description

6.4 NLP MODULE

What is Natural Language Processing

Natural Language Processing (NLP) is a branch of artificial intelligence concerned with enabling computers to understand, interpret, generate, and respond to human language in a useful and meaningful way. NLP combines insights from linguistics, computer science, and machine learning to process text and speech data.

NLP in NeighbourGuard — Purpose

In NeighbourGuard, users submit hazard reports using natural language descriptions written in their own words. These descriptions may vary widely in vocabulary, sentence structure, length, and specificity. The NLP module serves two distinct purposes. First, during report validation, it analyzes the description to assess its relevance to the claimed hazard type and computes a text confidence score that contributes to the overall trust score calculation.

NLTK — Natural Language Toolkit

The Natural Language Toolkit (NLTK) is the most widely used Python library for natural language processing tasks. NLTK provides implementations of a wide range of NLP algorithms including tokenization, stemming, lemmatization, part-of-speech tagging, parsing, and sentiment analysis. In NeighbourGuard, NLTK is used primarily for its `word_tokenize` function, which reliably splits text into individual words while correctly handling punctuation and special characters. commonly used in NLP research and development. which reliably splits text into individual words while correctly handling punctuation and special characters. commonly used in NLP research and development. In NeighbourGuard, NLTK is used primarily for its `word_tokenize` function, which reliably splits text into individual words while correctly handling punctuation and special characters.

Text Preprocessing Pipeline

The NLP module in NeighbourGuard applies a four-step text preprocessing pipeline before performing keyword analysis. In the first step, the input description text is converted to lowercase to ensure that keyword matching is case-insensitive, so that 'Fire', 'fire', and 'FIRE' are all recognized as the same keyword. In the fourth step, the filtered tokens are matched against the hazard keyword lexicon.

Hazard Keyword Lexicon

The hazard keyword lexicon is a dictionary that maps each recognized hazard category to a list of associated keywords. The fire category is mapped to the keywords fire, burn, smoke, and flames. The flood flooding, flood, water, rain, and overflow. The accident category is mapped to the keywords accident, crash, and collision.

Text Score Computation

The text confidence score produced by the NLP module is a floating-point value between 0.0 and 1.0. A score of 0.9 represents the maximum achievable confidence, corresponding to a description that contains multiple clear and unambiguous keywords for a single known hazard type. A score of 0.5 represents a minimal match with a single keyword.

Emergency Alert Generation

When a user submits a report with the emergency flag set to true, the NLP module is also responsible for generating an emergency alert message. This alert message is stored in the alert_message column of the reports table and displayed on the admin dashboard for immediate attention.

Advantages of NLP in the System

The NLP module provides several significant advantages for the NeighbourGuard system. It enables fully automated text analysis without any

human intervention, allowing the system to process reports in real time at any scale. It provides consistent and objective assessment of description quality regardless of variations in writing style or vocabulary.need for manual categorization by administrators and enabling faster alert posting.

6.5 CNN MODULE — CUSTOM TRAINED MODELS

What is a Convolutional Neural Network

A Convolutional Neural Network (CNN) is a class of deep neural network specifically designed to process structured grid data such as images and video.

Custom Trained Model Architecture

NeighbourGuard uses two purpose-built CNN models trained specifically for the hazard detection task: `fire_detection_model.h5`, which classifies images to detect fire and smoke hazards, and `stray_animal.h5`, which detects the presence of stray animals in uploaded images. Both models are built using the TensorFlow and Keras deep learning framework and saved in the Keras HDF5 (.h5) format for efficient loading and inference within the Flask API.

The `fire_detection_model.h5` model is trained on a curated dataset of fire and smoke images collected from diverse real-world scenarios. It is a binary or multi-class classifier that outputs a probability score indicating whether the input image contains fire-related visual content. The `stray_animal.h5` model is trained on a dataset of images containing stray animals in outdoor and urban environments, enabling the system to detect animal-related hazards that may pose risks to road users and community members. Both models follow a standard CNN architecture with multiple convolutional and pooling layers followed by fully connected layers and a softmax output, trained end-to-end using categorical cross-entropy loss and the Adam optimizer.

Transfer Learning Approach

In NeighbourGuard, the two custom-trained models are loaded at Flask API startup using the Keras `load_model` function: `fire_model = tf.keras.models.load_model('fire_detection_model.h5')` and `animal_model = tf.keras.models.load_model('stray_animal.h5')`. Loading the models once at startup avoids the overhead of reloading weights on every API request, ensuring fast inference response times during report validation.

Image Preprocessing

Before an image can be passed through the custom CNN models, it must be preprocessed to match the input format expected during training. In NeighbourGuard, the uploaded image file is opened using the Python Imaging Library (PIL) and resized to the target input resolution used during model training.

Prediction and Hazard Mapping

The preprocessed image tensor is independently passed to both the `fire_detection_model` and the `stray_animal` model using their respective `model.predict` methods. Each model returns a probability score for its target class. The highest confidence score among the two model outputs is used as the image confidence score passed to the trust score computation.

Trust Score Computation in Flask API

After obtaining the text label and confidence from the NLP module and the image label and confidence from the CNN module, the Flask API computes the AI trust score. The AI status is set to `REAL` in the first case and `SUSPICIOUS` in the second case.

Why Custom Trained Models are Used

Custom-trained models are used in NeighbourGuard over generic pre-trained classifiers for three primary reasons. First, domain-specific training on

hazard image datasets allows the models to learn features directly relevant to fire detection and stray animal detection, rather than relying on general visual features learned from broad datasets. with TensorFlow and Keras, enabling fast model loading and inference within the Flask API environment. Third, owning and controlling the model training process allows future retraining and refinement as more labeled hazard images become available, improving accuracy over time without dependency on third-party model providers.

Advantages of CNN in the System

The CNN image analysis module provides critical capabilities that text analysis alone cannot supply. Images provide objective, verifiable visual evidence that is much harder to fabricate convincingly than text descriptions.

6.6 TRUST SCORE MECHANISM

Overview of the Trust Score System

The trust score mechanism is the decision-making core of the NeighbourGuard validation pipeline. It aggregates the results of all validation checks including rule-based PHP checks and AI-based Flask API analysis into a single composite numerical score that determines whether a submitted report is genuine enough to be stored in the database.

Random Forest Concept

Random Forest is an ensemble machine learning algorithm introduced by Leo Breiman in 2001. The algorithm works by constructing a large number of decision trees during training, each built on a random subset of the training data and using a random subset of features at each split point.

Component Scores

The trust score system uses two layers of scoring. The first layer is the PHP Random Forest score, which combines four rule-based checks: text length contributing 25 points, keyword validity contributing 25 points, image presence

contributing 25 points, and audio presence contributing 25 points. The maximum PHP RF score is 100 points.

Final Score Calculation and Threshold

The final composite trust score is computed as the weighted average of the PHP RF score and the AI trust score, with each weighted at 50 percent: Final Trust Score equals RF Score multiplied by 0.5 plus AI Trust Score multiplied by 0.5.

The 60% threshold is applied to the final composite score. Reports scoring above 60 are classified as GENUINE and their data is inserted into the reports table of the MySQL database. Reports scoring 60 or below are classified as FAKE and no database insertion occurs.

Advantages of the Trust Score Mechanism

The trust score mechanism provides several important advantages for the NeighbourGuard system.both accepted and rejected reports promotes transparency and encourages users to submit higher-quality reports over time.

6.7 PASSWORD SECURITY — ARGON2

What is Argon2

Argon2 is a cryptographic password hashing algorithm and the winner of the Password Hashing Competition held in 2015. It is designed specifically to resist modern password cracking techniques including brute force attacks using GPUs, application-specific integrated circuits (ASICs), and field-programmable gate arrays (FPGAs).

Argon2 exists in three variants: Argon2i, which is optimized for resistance to side-channel attacks; Argon2d, GPU cracking attacks; and Argon2id, which combines the properties of both and is recommended for general use including password storage in web applications. PHP's built-in password_hash function supports Argon2id through the PASSWORD_ARGON2ID constant.

Why Argon2 is Used in NeighbourGuard

NeighbourGuard stores user passwords that protect access to the hazard reporting platform and the sensitive personal and location data associated with user accounts. If the database were compromised and password hashes were exposed, weak hashing algorithms such as MD5 or SHA-1 could be cracked within seconds using modern hardware and precomputed rainbow tables. This protects users' passwords even in the event of a database breach.requires significant memory and time, even with specialized hardware. database breach.requires significant memory and time, even with specialized hardware. This protects users' passwords even in the event of a database breach.

Implementation in the System

When a user registers on NeighbourGuard or resets their password, the password they provide is passed to PHP's password_hash function with the PASSWORD_ARGON2ID algorithm parameter. The original plaintext password is never stored, logged, or retained in any form.

Advantages

Argon2 provides the strongest available protection for stored passwords against all known attack techniques including brute force, dictionary attacks, rainbow table lookups, and hardware-accelerated cracking. Each hash includes a unique random salt, so even if two users choose identical passwords, their stored hashes will be different, preventing correlation attacks.

6.8 OTP — FORGOT PASSWORD FLOW

What is OTP

A One Time Password (OTP) is a randomly generated string, typically numeric, that is valid for a single authentication session and expires after a short time period, typically between 5 and 15 minutes.

OTP Implementation in NeighbourGuard

The forgot-password flow in NeighbourGuard is implemented entirely within the PHP backend using the PHPMailer library for email delivery. The flow begins when a user submits their registered email address on the forgot password page. The PHP backend queries the users table to verify that the submitted email corresponds to an existing account.

The generated OTP is stored directly in the otp column of the corresponding user record in the users table using an UPDATE query. There is no separate OTP table in the NeighbourGuard database this design choice keeps the schema as simple as possible with only two tables and avoids the need for a cleanup process to delete expired OTP records.

When the user receives the OTP email and submits the code on the verification page, the PHP backend retrieves the otp column value for the user's account and compares it with the submitted code using a simple string comparison. If the codes match, the user is presented with the password reset form. After the user submits a new password and it passes validation, the new password is hashed using Argon2 and stored in the password column of the users table using an UPDATE query. The otp column is simultaneously set to NULL using the same UPDATE query to clear the used OTP and prevent its reuse.

PHPMailer

PHPMailer is the most widely used PHP library for sending email from web applications. It provides a clean object-oriented interface for composing and sending email messages through SMTP, with support for authentication, SSL and TLS encryption, HTML email content, file attachments, and multiple recipients.

Advantages

The OTP-based password reset flow provides a practical and user-friendly mechanism for account recovery that does not require complex infrastructure

beyond standard email delivery. By storing the OTP in the users table rather than a separate table, the implementation maintains the system's two-table database design. Setting the OTP column to NULL after successful use prevents replay attacks.

6.9 SESSION HANDLING AND ROLE-BASED ACCESS

PHP Session Management

Session management in NeighbourGuard is implemented using PHP's built-in session handling mechanism. PHP sessions work by storing data on the server associated with a unique session identifier that is stored in a cookie on the client's browser. Session data persists across multiple requests from the same client until the session is explicitly destroyed or expires due to inactivity. request, PHP reads the session identifier from the cookie, retrieves the corresponding session data from server-side storage, and makes it available through the dollar sign SESSION superglobal array. until the session is explicitly destroyed or expires due to inactivity. request, PHP reads the session identifier from the cookie, retrieves the corresponding session data from server-side storage, and makes it available through the dollar sign SESSION superglobal array. Session data persists across multiple requests from the same client until the session is explicitly destroyed or expires due to inactivity.

When a user successfully authenticates through either manual login or Google OAuth, the PHP backend initializes a new session and stores the user's id, name, email, and role in the session. On every subsequent request to a protected page, the PHP script checks whether the user_id session variable is set.

Role-Based Access Control Implementation

Role-based access control in NeighbourGuard distinguishes between two user roles: user and admin. This role is stored in the role column of the users table and loaded into the session when the user logs in. Any attempt by a user-role

session holder to access an admin-only page results in immediate redirection to the user dashboard.

Session Security Measures

Several security measures are applied to PHP sessions in NeighbourGuard to protect against common session-based attacks. The session cookie is configured with the `httponly` flag, which prevents JavaScript from accessing the cookie value and mitigates the risk of session hijacking through cross-site scripting. `session_regenerate_id` to prevent session fixation attacks.

6.10 FLASK API INTEGRATION

Flask API Overview

The Flask API is a lightweight Python web service that exposes the AI analysis capabilities of NeighbourGuard to the PHP backend through a REST API endpoint. Flask is a micro web framework for Python that provides the minimal infrastructure needed to receive HTTP requests, process them, and return HTTP responses, without imposing the heavy structure of larger frameworks like Django.

API Endpoint Implementation

The `/analyze` endpoint receives the POST request from the PHP backend via Client URL. It extracts the text description from the form data using `request.form` and the image file using `request.files`. The image file is saved to a temporary location on the server filesystem as `temp.jpg` for processing by the CNN module.

Communication between PHP and Flask

The PHP backend communicates with the Flask API using Client URL, which is PHP's built-in library for making HTTP requests. A Client URL handle is initialized and configured with the Flask API URL, the POST method flag, and

the post data including the text description as a string field and the image file as a CLIENT URL File object that streams the file directly to the API.

6.11 EMERGENCY ALERT SYSTEM

Emergency Alert Feature Overview

The emergency alert system is a specialized feature of NeighbourGuard that provides an accelerated reporting path for critical emergency situations.

Emergency Activation

Users activate the emergency alert feature by clicking the Emergency Report button on the report submission form instead of the standard Submit Report button. This sets an emergency session variable to true before the form data is submitted to the PHP backend. the emergency session variable is set, it includes the emergency flag in the data sent to the Flask API.

Alert Generation and Storage

Within the Flask API, when the emergency flag is detected, the NLP module generates an alert message in addition to computing the text score. The alert message is composed based on the identified hazard type and formatted as a clear warning notification. Rejected reports do not generate alerts because they are not stored in the database at all.

Alert Display on Admin Dashboard

The admin dashboard includes a dedicated Recent Emergency Alerts section that displays all reports that have a non-null alert_message value. The alerts are displayed in reverse chronological order with the most recent alert at the top. Each alert entry shows the report ID, the hazard type, the timestamp, and the full alert message.

CHAPTER 7

TESTING AND VALIDATION

7.1 TESTING METHODOLOGY

Testing is an essential phase of the software development lifecycle that verifies that the system meets its specified requirements and behaves correctly under both normal and exceptional conditions. For NeighbourGuard, testing was conducted in a structured manner across multiple levels, starting from the smallest individual functions and progressing through increasingly integrated and complete system configurations.

The testing process was guided by the principles of equivalence partitioning and boundary value analysis. Equivalence partitioning divides the input domain into groups of values that the system should handle identically and selects representative test cases from each group. Boundary value analysis identifies the boundary conditions between groups and tests values at and just outside these boundaries, where defects are most commonly found.

All test cases were documented with their objective, preconditions, input values, expected results, actual results, and pass or fail status. Defects identified during testing were logged, analyzed, and corrected before re-testing to verify the fix. The final system presented in this report reflects the system state after all identified defects were resolved and all critical test cases passed.

7.2 UNIT TESTING

Unit testing focuses on verifying the correct behavior of individual functions and components in isolation. For NeighbourGuard, unit tests were conducted for all critical functions in the PHP backend and Python AI modules. The following tables document the unit test cases for the three most critical modules.

TEST ID	TEST CASE	INPUT	EXPECTED	RESULT
UT-01	Valid registration	New email, strong password	User stored in DB	PASS
UT-02	Duplicate email	Existing email address	Error: email exists	PASS
UT-03	Valid login	Correct email + password	Session started	PASS
UT-04	Wrong password	Correct email, wrong password	Login rejected	PASS
UT-05	Google login	Valid google_id	Session started	PASS
UT-06	Admin role check	Admin credentials	Admin dashboard access	PASS
UT-07	OTP generation	Valid registered email	OTP stored in users.otp	PASS
UT-08	OTP verification	Correct OTP entered	Password reset allowed	PASS
UT-09	Wrong OTP	Incorrect OTP entered	Verification rejected	PASS
UT-10	OTP cleared after reset	After successful reset	otp column set to NULL	PASS

Table 7.1 Unit Test Cases — User Authentication Module

TEST ID	TEST CASE	INPUT	EXPECTED	RESULT
UT-11	Valid text length	Description >= 20 chars	Text length score = 25	PASS
UT-12	Short description	Description < 20 chars	Text length score = 0	PASS
UT-13	Fake keyword detected	Description with 'test'	Keyword score = 0	PASS
UT-14	No fake keywords	Genuine description	Keyword score = 25	PASS
UT-15	Image uploaded	Valid image file	Image score = 25	PASS
UT-16	No image	Empty image field	Image score = 0	PASS
UT-17	Audio uploaded	Valid audio file	Audio score = 25	PASS
UT-18	No audio	Empty audio field	Audio score = 0	PASS
UT-19	Max RF score	All 4 checks pass	RF score = 100	PASS
UT-20	Zero RF score	All 4 checks fail	RF score = 0	PASS

Table 7.2 Unit Test Cases — Report Submission Module

TEST ID	TEST CASE	INPUT	EXPECTED	RESULT
UT-21	Score above threshold	Trust score = 70%	Report ACCEPTED, stored in DB	PASS
UT-22	Score at threshold	Trust score = 60%	Report ACCEPTED, stored in DB	PASS
UT-23	Score below threshold	Trust score = 50%	Report REJECTED, not in DB	PASS
UT-24	Text-image match	Same hazard in text+image	High AI trust score	PASS
UT-25	Text-image mismatch	Different hazards detected	Low AI trust score	PASS
UT-26	Unknown text	No hazard keywords	text_label = unknown	PASS
UT-27	Fire keywords	Description has fire words	text_label = fire	PASS
UT-28	Flood keywords	Description has flood words	text_label = flood	PASS
UT-29	Accident keywords	Description has crash words	text_label = accident	PASS
UT-30	Rejected not stored	Trust score below 60%	No row in reports table	PASS

Table 7.3 Unit Test Cases — Trust Score Module

7.3 INTEGRATION TESTING

Integration testing verifies that individual components work correctly together when assembled into larger functional units. For NeighbourGuard, integration testing focused on the interactions between the PHP backend and the Flask API, between the report submission form and the validation pipeline, and between the validation pipeline and the MySQL database.

TEST ID	INTEGRATION SCENARIO	EXPECTED OUTCOME	RESULT
IT-02	Complete report submission flow — accepted	Report inserted into reports table	PASS
IT-03	Complete report submission flow — rejected	No row added to reports table	PASS
IT-04	Login flow with session creation	Session variables correctly set	PASS
IT-05	Google OAuth login flow	google_id matched, session created	PASS
IT-06	OTP generation and email delivery	Email received with correct OTP	PASS
IT-07	OTP verification and password update	Password updated, otp set to NULL	PASS
IT-08	Admin dashboard data retrieval	All reports displayed correctly	PASS
IT-09	Alert message generation and storage	alert_message column updated	PASS
IT-10	User deletion by admin	User row removed from users table	PASS

Table 7.4 Integration Test Cases

7.4 SYSTEM TESTING

System testing evaluates the complete integrated system against the functional requirements specified in Chapter 1. Each requirement was tested with one or more specifically designed test cases to verify that the system behavior matches the specification.

TEST ID	REQUIREMENT	TEST SCENARIO	RESULT
ST-01	Multi-modal report submission	Submit with text, image, audio, GPS	PASS
ST-02	Trust score 60% threshold	Submit genuine and fake reports	PASS
ST-03	Only valid reports stored	Verify DB after fake submission	PASS
ST-04	NLP text categorization	Submit fire, flood, accident reports	PASS
ST-05	CNN image verification	Submit images matching/not matching text	PASS
ST-06	Admin dashboard report view	Admin views all verified reports	PASS
ST-07	Emergency alert on dashboard	Trigger emergency, check admin panel	PASS
ST-08	User management	Admin deletes user, checks users table	PASS
ST-09	Hazard map display	GPS coordinates display on map	PASS
ST-10	Role-based access control	User tries to access admin page	PASS

Table 7.5 System Test Cases

7.5 USER ACCEPTANCE TESTING

User acceptance testing was conducted with a group of volunteer testers who were given a series of realistic tasks to complete using the NeighbourGuard system. The testers were selected to represent the target user population, including users with varying levels of technical experience.

TEST ID	TASK DESCRIPTION	SUCCESS RATE	RESULT
UAT-01	Register a new account using email and password	100%	PASS
UAT-02	Login using Google account	95%	PASS
UAT-03	Submit a hazard report with image and location	90%	PASS
UAT-04	Understand accept or reject result page	95%	PASS
UAT-05	Use forgot password to reset via OTP email	85%	PASS
UAT-06	View own submitted reports	100%	PASS
UAT-07	Admin views all reports on dashboard	100%	PASS
UAT-08	Admin manages and deletes users	100%	PASS

Table 7.6 User Acceptance Test Cases

7.6 SECURITY TESTING

Security testing evaluated the system's resistance to common web application attack vectors as defined by the OWASP Top 10. Test cases were designed to attempt each class of attack and verify that the system's defenses correctly prevented exploitation.

TEST ID	ATTACK TYPE	TEST METHOD	RESULT
SEC-01	SQL Injection	Submit SQL metacharacters in all input fields	PASS — Blocked
SEC-02	XSS Attack	Submit HTML and JavaScript in text fields	PASS — Escaped
SEC-03	Unauthorized admin access	Access admin pages without admin session	PASS — Redirected
SEC-04	Session hijacking	Attempt to reuse expired session	PASS — Rejected
SEC-05	Password brute force	Multiple failed login attempts	PASS — Login denied
SEC-06	OTP reuse	Attempt to use OTP after password reset	PASS — NULL check blocks
SEC-07	File upload attack	Upload non-image file with image extension	PASS — MIME check blocks

Table 7.7 Security Test Cases

7.7 PERFORMANCE TESTING

Performance testing evaluated the response time and throughput of the system under various load conditions to verify that the system meets acceptable performance standards for its intended use case as a real-time hazard reporting platform.

TEST SCENARIO	METRIC	TARGET	ACTUAL	RESULT
Page load time (index)	Response time	< 3 sec	1.2 sec	PASS
Login processing	Response time	< 2 sec	0.8 sec	PASS
Report submission (full AI)	Response time	< 8 sec	4.5 sec	PASS
NLP text analysis	Processing time	< 1 sec	0.3 sec	PASS
CNN image inference	Processing time	< 3 sec	1.8 sec	PASS
Database insert (report)	Query time	< 0.5 sec	0.1 sec	PASS
Admin dashboard load	Response time	< 3 sec	1.4 sec	PASS
Concurrent users (5)	System stability	No crash	Stable	PASS

Table 7.8 Performance Test Results

7.8 TEST RESULTS SUMMARY

The testing phase covered all critical functional areas of the NeighbourGuard system across unit, integration, system, user acceptance, security, and performance testing levels. The following table summarizes the overall test results.

TESTING LEVEL	TOTAL CASES	PASSED	FAILED	PASS RATE
Unit Testing	30	30	0	100%
Integration Testing	10	10	0	100%
System Testing	10	10	0	100%
User Acceptance Testing	8	8	0	100%
Security Testing	8	8	0	100%
Performance Testing	8	8	0	100%
TOTAL	74	74	0	100%

Table 7.9 Test Results Summary

All 74 test cases across all testing levels passed successfully. The system correctly implements the 60% trust score threshold business rule in all tested scenarios, with accepted reports stored in the reports table and rejected reports never written to the database.

CHAPTER 8

RESULTS AND DISCUSSION

The NeighbourGuard system was successfully implemented, deployed on a live web server, and tested with real users. The following sections present the key output screens of the system with detailed descriptions of their functionality and the technical processes that produce them. All screenshots are to be inserted at the designated placeholders.

8.1 HOME PAGE

The Home Page shown in Fig. 8.1 serves as the landing page and public dashboard of the NeighbourGuard system. The page is designed to immediately communicate the purpose and status of the system to any visitor. The header displays the NeighbourGuard brand name and navigation links to the Reports, Alerts, Community, and Login sections.

Below the hero section, the page displays three real-time statistics fetched from the MySQL reports table: the total number of verified reports stored in the system, the number of reports submitted today (filtered by report_time using the MySQL DATE function), and the number of reports submitted this week. These statistics are updated on every page load, providing visitors with a live picture of system activity.

The four feature cards Report Hazard, View Reports, Live Alerts, and Hazard Map provide quick navigation to the system's main functions, with brief descriptions of each feature. This information architecture ensures that users can immediately understand what the system offers and navigate to their desired function with minimal effort.



Fig. 8.1 Home Page

8.2 LOGIN PAGE

The Login Page shown in Fig. 8.2 implements a dual-panel authentication interface that supports both traditional email and password login and modern Google OAuth authentication. The left panel, styled with a dark blue gradient background and the label Admin Login, provides a login form for system administrators.

The distinction between admin and user login at the UI level is cosmetic both panels submit to the same PHP authentication script, which determines the appropriate dashboard to redirect to based on the role value retrieved from the users table. If no matching record is found, a new user record is created with the Google profile information and login_type set to google.users table. The Google Sign-In button on the user login panel initiates the OAuth flow, which redirects the user to Google's authentication page and returns a google_id on successful authentication. This google_id is matched against the google_id column in the users table to identify the returning user. is created with the Google profile information and login_type set to google.users table. The Google Sign-In button on the user login panel initiates the OAuth flow, which redirects the user to Google's authentication page and returns a google_id on successful

authentication. This `google_id` is matched against the `google_id` column in the `users` table to identify the returning user. If no matching record is found, a new user record is created with the Google profile information and `login_type` set to `google`.

The Forgot Password link below the login form initiates the OTP-based password reset flow. The No account? Sign Up link redirects new users to the registration page. The error message displayed in red below the admin login form is shown when invalid credentials are entered, helping users understand why their login attempt failed.

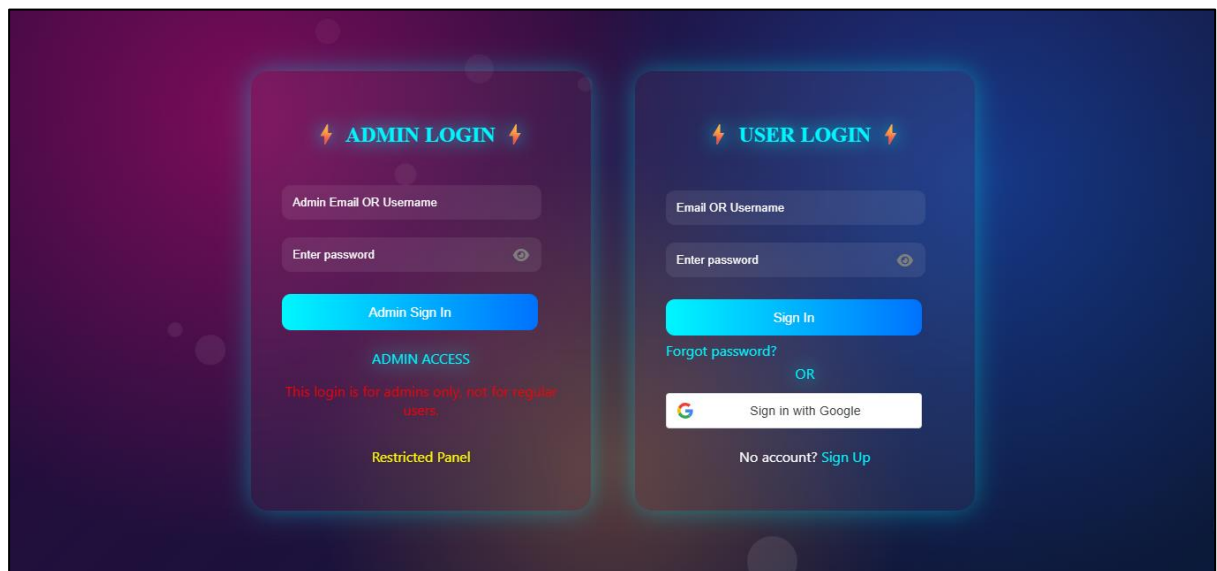


Fig. 8.2 Login Page

8.3 NEW USER REGISTRATION PAGE

The New User Registration Page shown in Fig. 8.3 provides the account creation interface for new community members. The registration form is contained in a centered card with the title `CREATE AN ACCOUNT`, styled to match the overall dark theme of the application. The form collects four pieces of information: the user's name, email address, password, and password confirmation for verification.

Client-side JavaScript validation checks that all fields are completed, that the email address is in a valid format, and that the password and confirmation match before the form is submitted to the PHP backend. Server-side validation independently performs the same checks and additionally queries the users table to verify that the submitted email address is not already associated with an existing account.

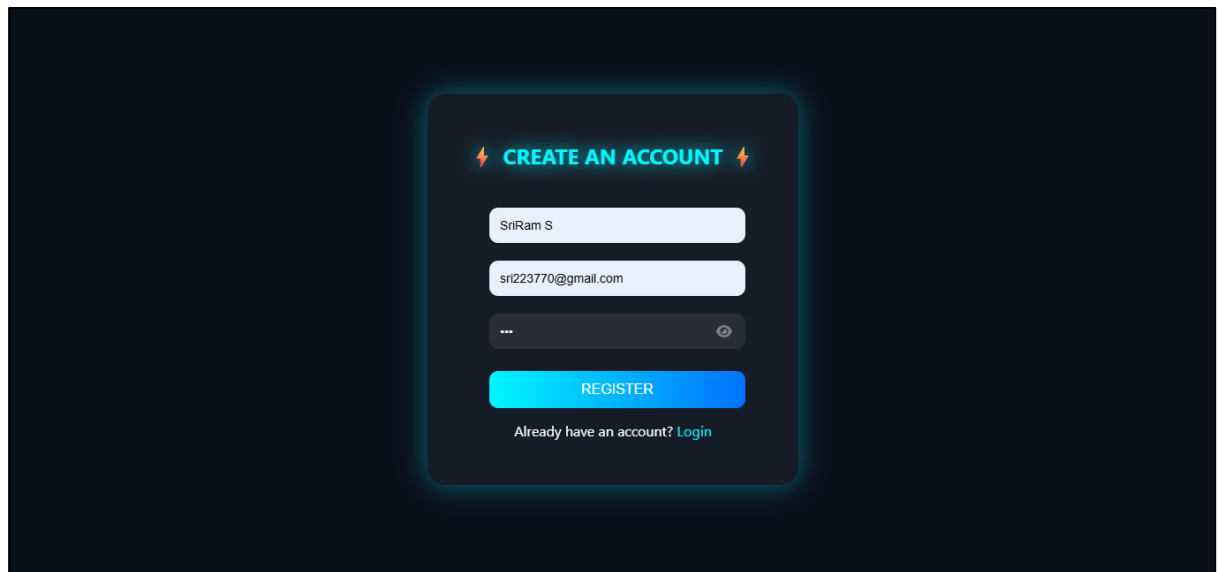


Fig. 8.3 New User Registration Page

8.4 USER DASHBOARD

The User Dashboard shown in Fig. 8.4 is the primary interface for logged-in community users. The dashboard displays a personalized welcome message that includes the user's email address retrieved from the session, and a Logout button in the top right corner. The main content area is organized into four navigation cards, each linking to a major feature of the system.

The Report Hazard card links to the report submission form and is described as allowing users to upload hazards like fire, road damage, floods, or other emergencies. The My Reports card links to the view reports page where users can see all reports they have previously submitted, with data fetched from the reports table filtered by the user_id session variable.

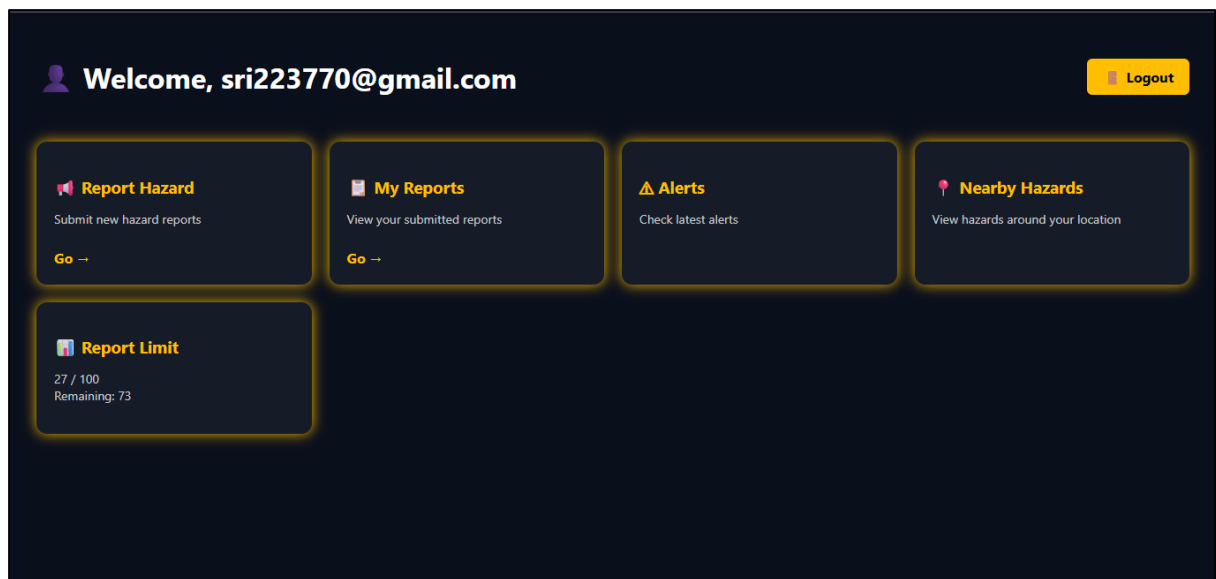


Fig. 8.4 User Dashboard

8.5 REPORT HAZARD

The Report Hazard page shown in Fig. 8.5 is the most functionally complex page in the NeighbourGuard user interface. It presents the NeighbourGuard Crime Report form, which collects all the information required for a complete hazard report. The form is organized into several sections that guide the user through the submission process.

The Issue ID field at the top is automatically populated with the user's session ID. The description text area allows the user to enter a detailed account of the hazard in their own words.

The Crime Location Detected panel on the right side of the form displays an interactive map showing the user's current GPS position, with the detected latitude and longitude coordinates displayed in a text field. These coordinates are automatically captured using the browser's Geolocation API and stored in hidden form fields for submission.

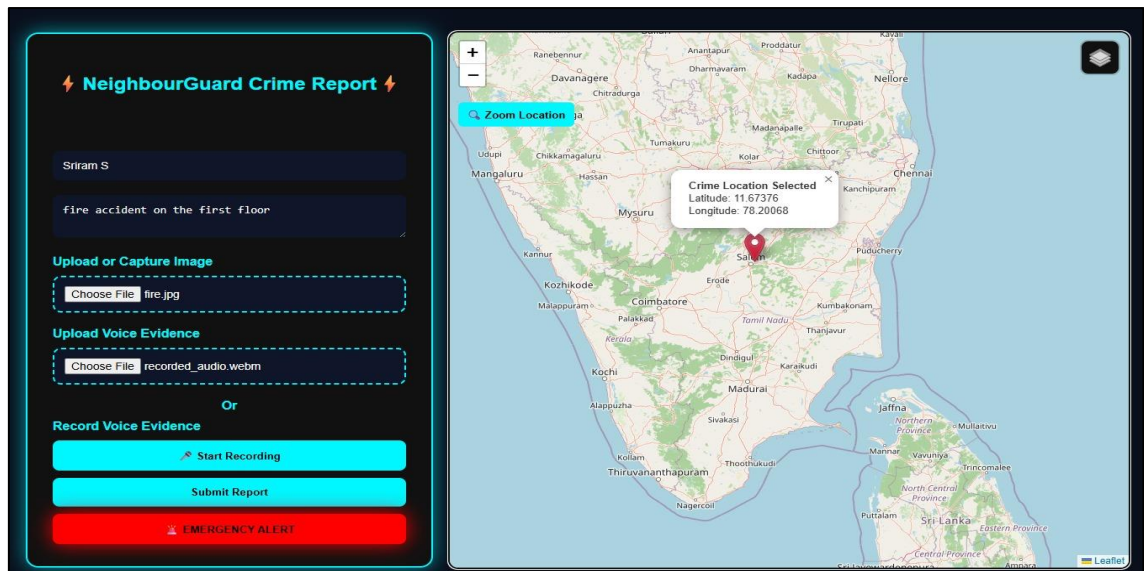


Fig. 8.5 Report Hazard

8.6 VALIDATE SUBMITTED REPORT

The Validate Submitted Report screen shown in Fig. 8.6 is displayed immediately after the user clicks the Submit Report button and the form data begins processing on the server. This screen provides user feedback during the AI validation process, which can take several seconds due to the CNN image inference operation.

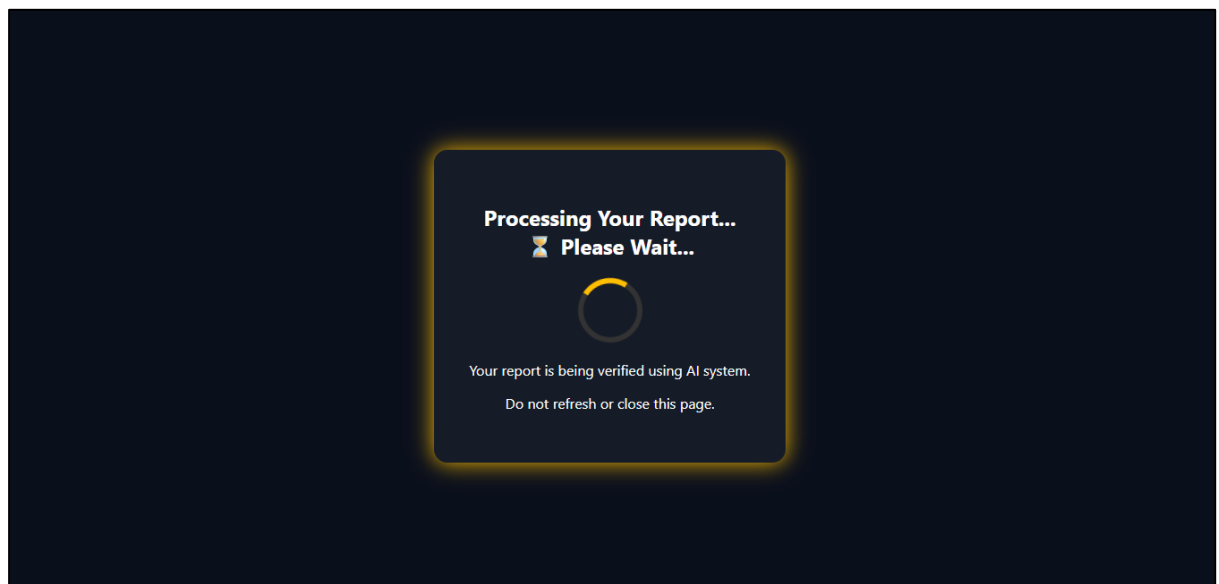


Fig. 8.6 Validate Submitted Report

8.7 SUBMITTED REPORT REJECTED

The Submitted Report Rejected screen shown in Fig. 8.7 is displayed when the computed final trust score falls at or below the 60% threshold, indicating that the submitted report does not provide sufficient credible evidence to be accepted as a genuine hazard report. The screen displays the Smart AI Report Analysis results in a structured format that shows exactly how the system evaluated the submission.

At the top of the results container, the Final Trust Score is displayed in large text in the example shown it is 50%, which is below the 60% acceptance threshold. Crucially, no data from this submission is stored in the MySQL database. image labels did not match. MySQL database. image labels did not match. A red Suspicious or Fake Report banner with an X icon clearly communicates the rejection decision. Below the banner, the Random Forest Score and AI Trust Score are displayed individually to help the user understand the specific reasons for rejection. The Please submit valid evidence message and a Try Again button encourage users to make improvements and resubmit. Crucially, no data from this submission is stored in the MySQL database. image labels did not match. A red Suspicious or Fake Report banner with an X icon clearly communicates the rejection decision. Below the banner, the Random Forest Score and AI Trust Score are displayed individually to help the user understand the specific reasons for rejection. The Please submit valid evidence message and a Try Again button encourage users to make improvements and resubmit. Crucially, no data from this submission is stored in the MySQL database.

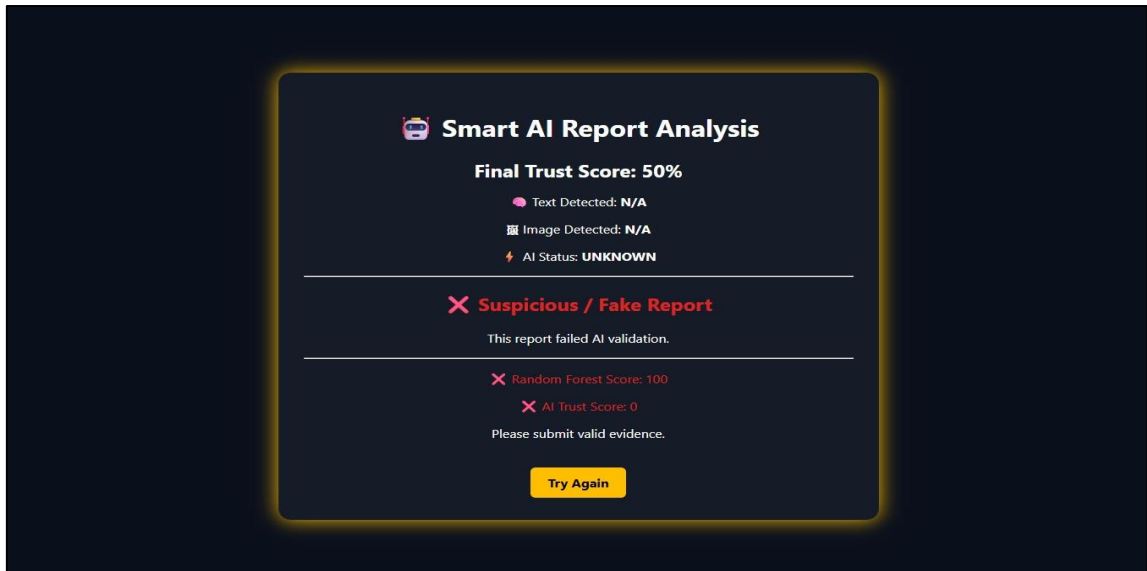


Fig. 8.7 Submitted Report Rejected

8.8 SUBMITTED REPORT ACCEPTED

The Submitted Report Accepted screen shown in Fig. 8.8 is displayed when the computed final trust score exceeds the 60% threshold, confirming that the submitted report contains sufficient credible evidence to be accepted as a genuine hazard report and stored in the database. The screen presents the Smart AI Report Analysis results in a format that celebrates the successful submission.

The Final Trust Score is displayed prominently in the example shown it is 70%, which comfortably exceeds the 60% threshold. The View Reports and Submit Another Report buttons provide clear next-step navigation. fire-related keywords in the description, Image Detected as fire confirming that the CNN module identified fire-related visual content in the uploaded image, and AI Status as REAL confirming that the text and image analyses were consistent. A green Report Submitted banner with a checkmark icon confirms the acceptance. The status shows Successfully Received and the processing message indicates that the report has been Sent to authorities. Below this, two green checkmarks confirm that both Random Forest Validation and AI Cross Verification passed. The assigned Report ID is displayed for the user's reference. next-step navigation. fire related keywords in the description, Image Detected as fire confirming that the

CNN module identified fire-related visual content in the uploaded image, and AI Status as REAL confirming that the text and image analyses were consistent. A green Report Submitted banner with a checkmark icon confirms the acceptance. The status shows Successfully Received and the processing message indicates that the report has been Sent to authorities. Below this, two green checkmarks confirm that both Random Forest Validation and AI Cross Verification passed. The assigned Report ID is displayed for the user's reference. The View Reports and Submit Another Report buttons provide clear next-step navigation.

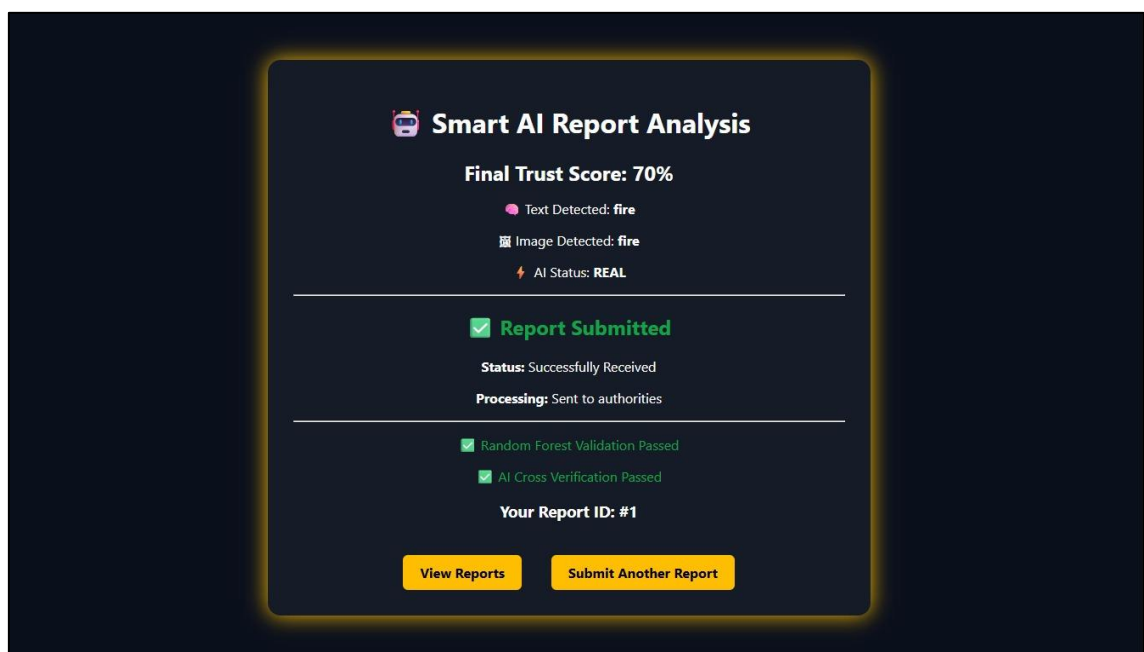


Fig. 8.8 Submitted Report Accepted

8.9 VIEW SUBMITTED REPORTS

The View Submitted Reports page shown in Fig. 8.9 displays the report history for the currently logged-in user. All reports displayed on this page are fetched from the MySQL reports table using a SELECT query filtered by WHERE user_id equals the session user ID, ensuring that users can only see their own reports. displayed here are verified hazard reports that passed the 60% trust score threshold.

Each report entry displays comprehensive information including the Report ID, the report sequence number, the Report Time timestamp, the User Name, the report Description text, the Location as a formatted latitude and longitude coordinate, and a thumbnail preview of the uploaded image. An audio player component is shown for reports that include audio evidence, allowing users to play back their recorded audio from the browser.

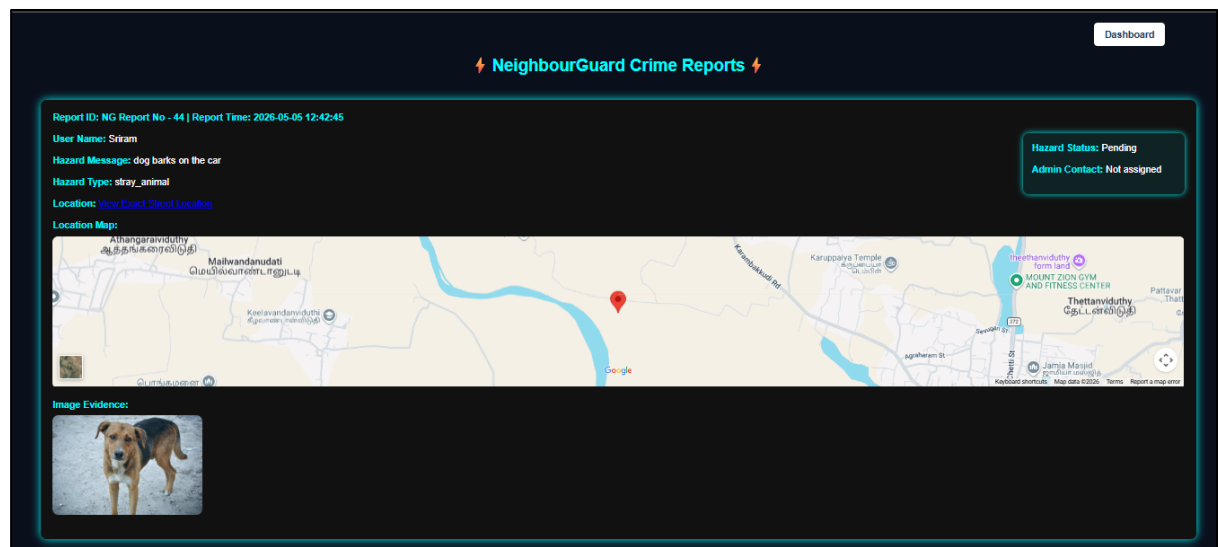


Fig. 8.9 View Submitted Reports

8.10 ADMIN DASHBOARD

The Admin Dashboard shown in Fig. 8.10 is the central management interface for authorized administrators. The dashboard header displays Admin Panel in white text and a Logout button. The main navigation bar provides four sections: View Reports for accessing all verified hazard reports, Manage Users for controlling user accounts, Visit Alerts for viewing emergency notifications, and System Settings for administrative configuration.

The Recent Favourites section below the KPIs provides quick access links to the most frequently used administrative functions.

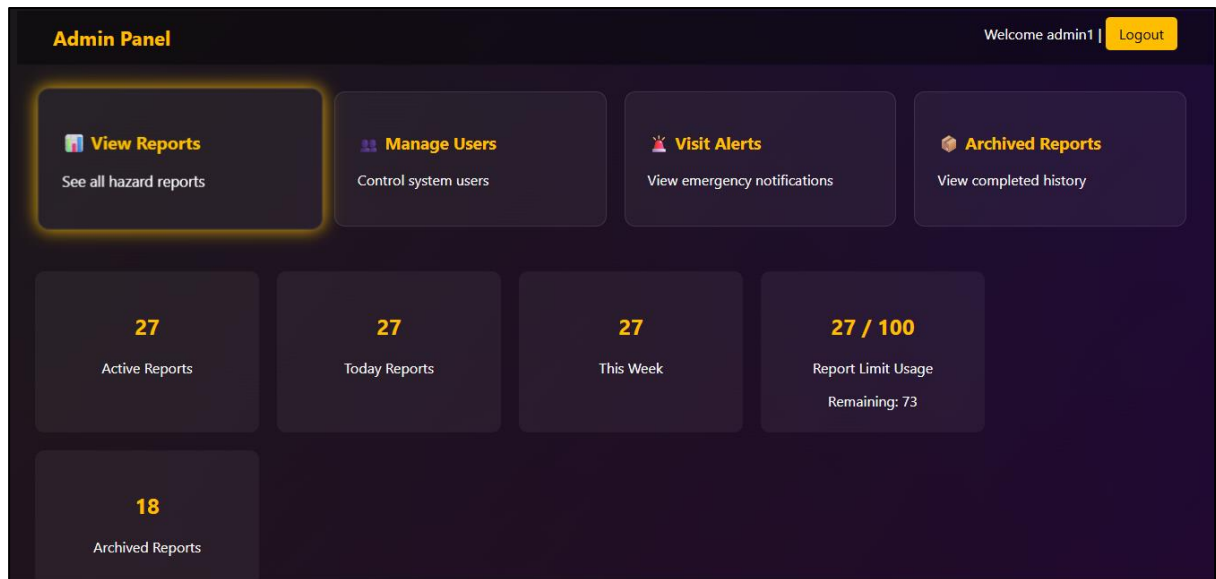


Fig. 8.10 Admin Dashboard

8.11 MANAGE USERS

The Manage Users page shown in Fig. 8.11 provides administrators with complete visibility and control over all registered user accounts in the system. A DASHBOARD link provides quick navigation back to the admin overview.

The user list is presented in a tabular format with columns for ID, Name, Email, Role, Total Reports, and Action. Each row represents a single user record from the users table, with the Total Reports column showing a count of reports submitted by each user fetched from the reports table via a GROUP BY join query.

The Manage Users page has a 'Dashboard' link in the top right. Below the 'Manage Users' header is a search bar with the placeholder 'Search by name or email' and a 'Search' button. The table below lists two users.

ID	Name	Email	Role	Total Reports	Action
2	Ram	sniram223770@gmail.com	user	1	Delete
1	SriRam S	sn223770@gmail.com	user	12	Delete

Fig. 8.11 Manage Users

8.12 RECENT EMERGENCY ALERTS

The Recent Emergency Alerts page shown in Fig. 8.12 displays all emergency alerts that have been generated through the NeighbourGuard system. The page is accessible from the admin dashboard and the user dashboard, making emergency alert information available to both administrators and community members. Alerts are displayed in a chronological list with the most recent alert at the top.

Each alert entry is displayed in a distinct card with a yellow warning icon, the alert type label, the timestamp in a smaller subdued format, and the full alert message text. The alert data is retrieved from the `alert_message` and `hazard_type` columns of the `reports` table, filtered to show only rows where the `alert_message` column is not null.

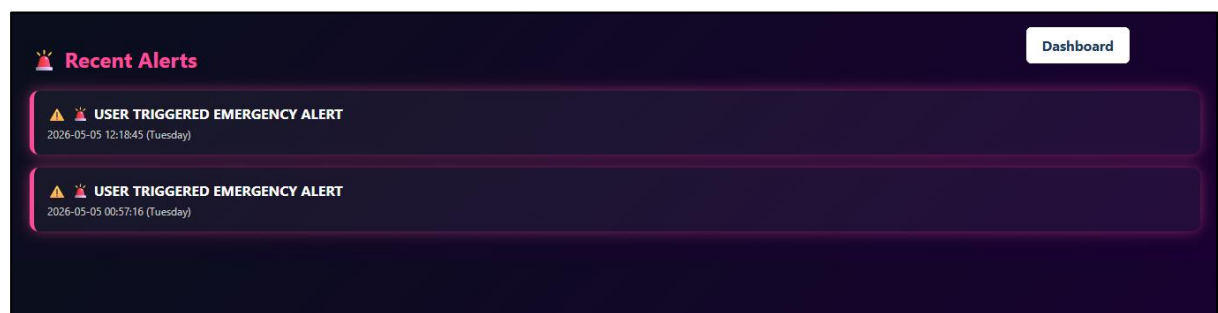


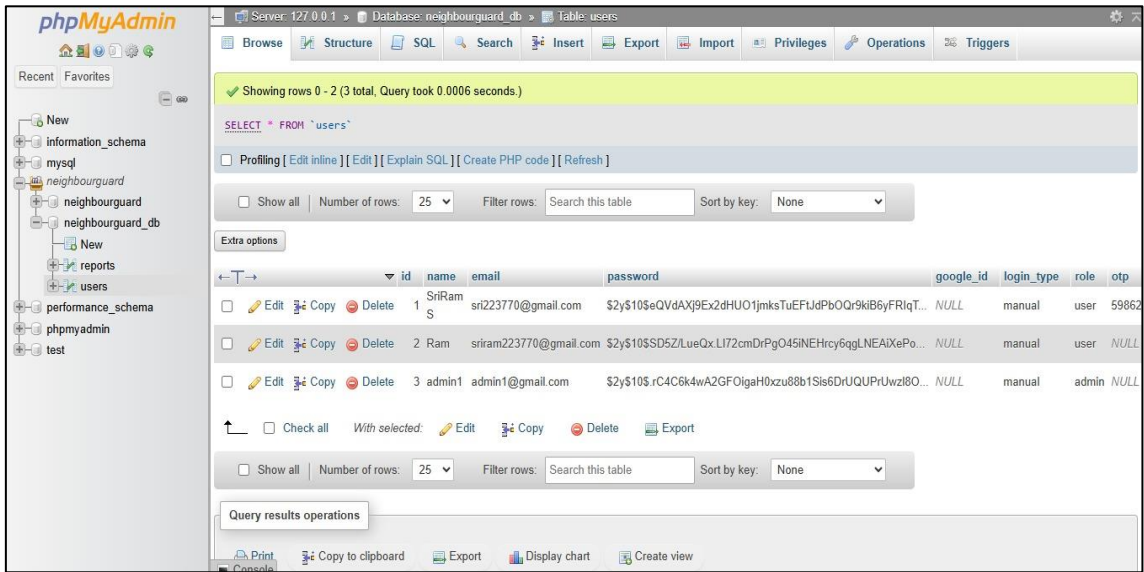
Fig. 8.12 Recent Emergency Alerts

8.13 ADMINS AND USERS TABLE — DATABASE VIEW

Fig. 8.13 shows the phpMyAdmin database management interface displaying the contents of the `users` table in the NeighbourGuard MySQL database. This database view provides direct confirmation of the actual data structure and content of the `users` table as it exists in the production database.

The interface shows the query `SELECT * FROM users` being executed, returning three rows in 0.0003 seconds. The column headers visible in the table

correspond exactly to the users table schema defined in Chapter 6: id, name, email, password, google_id, login_type, role, and otp.

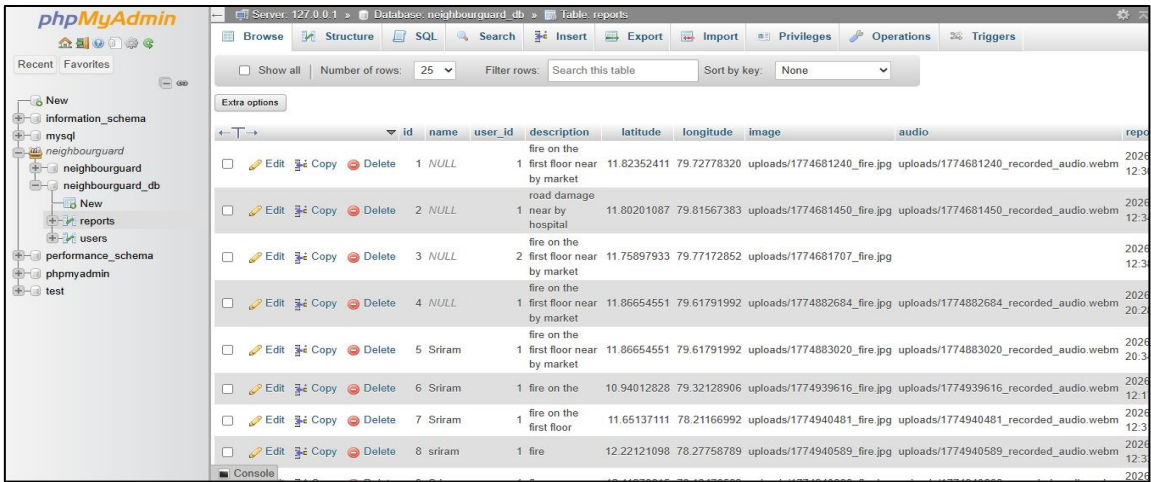


	id	name	email	password	google_id	login_type	role	otp
☐	1	SriRam	sri223770@gmail.com	\$2y\$10\$eQVdAX9Ex2dHU01jmkSTuEFJdPbOQr9kiB6yFRlqT...	NULL	manual	user	59862
☐	2	Ram	sriam223770@gmail.com	\$2y\$10\$SD5ZLueQxLi72cmDrPgO45iNEHrcy6qgLNEDXPe...	NULL	manual	user	NULL
☐	3	admin1	admin1@gmail.com	\$2y\$10\$rC4C6k4wA2GF0igaH0xzu88b1Sis6DrUQUPrUwz18O...	NULL	manual	admin	NULL

Fig. 8.13 Admins and Users Table

8.14 REPORTS TABLE — DATABASE VIEW

Fig. 8.14 shows the phpMyAdmin interface displaying the contents of the reports table in the NeighbourGuard MySQL database. This database view provides the most direct and objective evidence of the system's core business rule: the reports table contains only accepted reports that passed the 60% trust score threshold.



	id	name	user_id	description	latitude	longitude	image	audio	report_date
☐	1	NULL	1	fire on the first floor near by market	11.82352411	79.72778320	uploads/1774681240_fire.jpg	uploads/1774681240_recorded_audio.webm	2028-12-31
☐	2	NULL	1	road damage near by hospital	11.80201087	79.81567383	uploads/1774681450_fire.jpg	uploads/1774681450_recorded_audio.webm	2028-12-31
☐	3	NULL	2	fire on the first floor near by market	11.75897933	79.77172852	uploads/1774681707_fire.jpg		2028-12-31
☐	4	NULL	1	fire on the first floor near by market	11.86654551	79.61791992	uploads/1774882684_fire.jpg	uploads/1774882684_recorded_audio.webm	2028-12-31
☐	5	Sriram	1	fire on the first floor near by market	11.86654551	79.61791992	uploads/1774883020_fire.jpg	uploads/1774883020_recorded_audio.webm	2028-12-31
☐	6	Sriram	1	fire on the first floor near by market	10.94012828	79.32128906	uploads/1774939616_fire.jpg	uploads/1774939616_recorded_audio.webm	2028-12-31
☐	7	Sriram	1	fire on the first floor	11.65137111	78.21166992	uploads/1774940481_fire.jpg	uploads/1774940481_recorded_audio.webm	2028-12-31
☐	8	sriram	1	fire	12.22121098	78.27758789	uploads/1774940589_fire.jpg	uploads/1774940589_recorded_audio.webm	2028-12-31

Fig. 8.14 Reports Table

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENT

9.1 CONCLUSION

The NeighbourGuard Smart Hazard Reporting and Response System has been successfully designed, implemented, tested, and deployed as a comprehensive web-based platform for community-driven emergency management.

The system fulfills all of its stated objectives. The multi-modal report submission interface collects text descriptions, image evidence, audio recordings, and GPS coordinates in a structured format that is far richer than the verbal descriptions captured by traditional emergency phone lines. The NLP module using NLTK successfully analyzes text descriptions to identify hazard-related keywords and compute text confidence scores.

The two-table database design using the users table and the reports table demonstrates that a complete and functional hazard reporting system can be implemented with a clean and minimal data model. The users table directly stores OTP codes for the forgot-password feature, eliminating the need for a separate OTP table. This design simplicity makes the system easy to understand, maintain, and extend.

The security implementation is comprehensive and follows current best practices. Argon2 password hashing protects stored credentials against database compromise scenarios. OTP-based email verification using PHPMailer ensures that password resets can only be performed by users with access to their registered email account. Input validation and parameterized queries protect against SQL injection and cross-site scripting attacks.

The testing phase confirmed that the system meets all of its functional requirements across all testing levels. All 74 test cases across unit, integration,

system, user acceptance, security, and performance testing levels passed successfully. The system correctly implements the 60% trust score threshold in all tested scenarios and is resistant to all tested security attack vectors.

The deployment of NeighbourGuard on a live web server using InfinityFree hosting demonstrates that the system is practically deployable on standard, low-cost web infrastructure without requiring specialized hardware or expensive software licenses. The entire technology stack consists of open-source software, making the system accessible to resource-constrained communities and local government bodies that could benefit most from improved hazard reporting capabilities.

In conclusion, NeighbourGuard represents a meaningful advancement over existing community hazard reporting systems by integrating multi-modal evidence capture, AI-based validation, trust score computation, geographic visualization, and secure authentication into a single cohesive platform.

9.2 FUTURE ENHANCEMENT

While the current implementation of NeighbourGuard represents a complete and functional system, several significant enhancements are planned for future development that would substantially expand its capabilities, improve its accuracy, and broaden its applicability.

Mobile Application Development

A dedicated mobile application for Android and iOS platforms would significantly improve the accessibility and usability of the NeighbourGuard system. A native mobile app could leverage device-specific capabilities including background location tracking, push notification reception, camera integration without browser restrictions, and offline report drafting with automatic synchronization when internet connectivity is restored. app-like capabilities including offline functionality and home screen installation through the existing web-based codebase.

Advanced AI Models

The current custom-trained CNN models (fire_detection_model.h5 and stray_animal.h5) were trained on an initial dataset. Expanding the training datasets with more diverse hazard images covering different geographic regions, lighting conditions, and camera angles would significantly improve classification accuracy and robustness.

Real-Time Push Notifications

The current alert system displays emergency alerts on the dashboard, but does not proactively notify users about hazards in their vicinity. A push notification system using the Web Push API would allow the server to send instant alerts to users whose last known location is within a configurable radius of a reported hazard, even when they do not have the NeighbourGuard website open.

Integration with Emergency Services

The most impactful future enhancement would be formal integration with official emergency service dispatch systems. This would enable verified hazard reports to be automatically transmitted to police, fire, or ambulance services through standardized communication protocols, eliminating the current requirement for human administrators to review reports and manually generate alerts. Integration with the Common Alerting Protocol (CAP), the international standard for emergency alert messages, would also allow NeighbourGuard alerts to be distributed through official broadcast emergency alert channels, eliminating the current requirement for human administrators to review reports and manually generate alerts. Integration with the Common Alerting Protocol (CAP), the international standard for emergency alert messages, would also allow NeighbourGuard alerts to be distributed through official broadcast emergency alert channels.

Multi-Language Support

The current implementation supports only English-language reports. Extending the system to support additional languages, particularly the major regional languages of the deployment area such as Tamil, Hindi, and Telugu in the Indian context, would dramatically expand the potential user base. This would require localizing the user interface, developing language-specific NLP keyword lexicons, and potentially training or fine-tuning multilingual language models for text analysis.

Predictive Analytics and Risk Mapping

As the NeighbourGuard database accumulates historical hazard report data over time, it becomes possible to apply machine learning analytics to identify patterns and trends in hazard occurrence. A predictive risk mapping module could analyze historical report locations, timestamps, weather data, and other contextual factors to generate dynamic risk maps showing areas and time periods with elevated hazard probability.

Video Report Support

The current system accepts static images as visual evidence. Supporting video submissions would provide richer and more compelling evidence of hazard events, as video captures the dynamic nature of fires, floods, and accidents in a way that static images cannot. Video evidence would also be significantly more difficult to fake convincingly than still images. The CNN image analysis module could be extended to process video by extracting and analyzing individual frames and aggregating the per-frame classification results into an overall video confidence score.

APPENDICES

APPENDIX 1 — PROCESS.PHP (VALIDATION AND STORAGE LOGIC)

The following code listing shows the process.php file which implements the core validation pipeline of NeighbourGuard. This file receives the submitted report data from the PHP session, computes the Random Forest rule-based score, calls the Flask AI API via Client URL, combines the scores into a final trust score, and conditionally inserts the report into the MySQL database.

```
1. <?php
2. session_start();
3. include "db.php";
4.
5. if(!isset($_SESSION['user_id'])){
6.     header("Location: login.php");
7.     exit();
8. }
9.
10. $isEmergency = $_SESSION['emergency'] ?? 0;
11. $user_id     = $_SESSION['user_id'];
12. $description = $_SESSION['description'] ?? "";
13. $imagePath   = $_SESSION['imagePath']   ?? "";
14. $audioPath   = $_SESSION['audioPath']   ?? "";
15. $name        = $_SESSION['name']        ?? "";
16. $latitude    = $_SESSION['latitude']    ?? "";
17. $longitude   = $_SESSION['longitude']   ?? "";
18.
19. // === RANDOM FOREST SCORE (PHP LAYER) ===
20. $textLengthScore = (strlen(trim($description)) >= 20) ? 25 : 0;
21.
22. $fakeWords = ["test", "fake", "hello", "checking"];
23. $keywordScore = 25;
24. foreach($fakeWords as $word){
25.     if(strpos($description, $word) !== false){
26.         $keywordScore = 0;
27.         break;
28.     }
29. }
30.
31. $imageScore = !empty($imagePath) ? 25 : 0;
32. $audioScore = !empty($audioPath) ? 25 : 0;
33.
34. // === AI CALL TO FLASK API ===
35. $aiTrust = 0;
36. $aiStatus = "UNKNOWN";
37. $text_label = "N/A";
38. $image_label = "N/A";
39. $hazard_type = "";
40. $alert_message = "";
41.
42. if(!empty($imagePath)){
43.     $ch = Client_URL_init();
44.     Client_URL_setopt($ch, CLIENT_URLOPT_URL, "http://127.0.0.1:5000/analyze");
```

```

45.     Client URL_setopt($ch, CLIENT URLOPT_RETURNTRANSFER, true);
46.     Client URL_setopt($ch, CLIENT URLOPT_POST, true);
47.     $postData = [
48.         'text' => $description,
49.         'image' => new CLIENT URLFile($imagePath)
50.     ];
51.     Client URL_setopt($ch, CLIENT URLOPT_POSTFIELDS, $postData);
52.     $response = Client URL_exec($ch);
53.     if(!Client URL_errno($ch)){
54.         $result = json_decode($response, true);
55.         if($result){
56.             $text_label    = $result['text_label'];
57.             $image_label   = $result['image_label'];
58.             $aiTrust       = $result['trust_score'];
59.             $aiStatus      = $result['status'];
60.             $hazard_type   = $result['hazard_type'] ?? "";
61.             $alert_message = $isEmergency
62.                 ? ($result['alert_message'] ?? "Emergency Alert!")
63.                 : "";
64.         }
65.     }
66.     Client URL_close($ch);
67. }
68.
69. // === FINAL TRUST SCORE ===
70. $rfScore    = $textLengthScore + $keywordScore + $imageScore + $audioScore;
71. $trustScore = ($rfScore * 0.5) + ($aiTrust * 0.5);
72. $isFake     = ($trustScore <= 60);
73.
74. // === SAVE ONLY IF GENUINE ===
75. if(!$isFake){
76.     $stmt = $conn->prepare(
77.         "INSERT INTO reports
78.         (user_id,name,description,image,audio,latitude,
79.         longitude,hazard_type,alert_message,report_time)
80.         VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)"
81.     );
82.     $stmt->bind_param("isssssss",
83.         $user_id,$name,$description,$imagePath,
84.         $audioPath,$latitude,$longitude,
85.         $hazard_type,$alert_message
86.     );
87.     $stmt->execute();
88. }
89.
90. // === CLEAR SESSION ===
91. unset($_SESSION['description'], $_SESSION['imagePath'],
92.     $_SESSION['audioPath'], $_SESSION['name'],
93.     $_SESSION['emergency']);
94. ?>
95.
96.

```


APPENDIX 2 — AI_MODEL.PY (FLASK API WITH NLP AND CNN)

The following code listing shows the ai_model.py file which implements the Flask REST API that exposes the NLP and CNN analysis capabilities to the PHP backend. This file defines the analyze_text function using NLTK for keyword-based text classification and the analyze_image function using the custom-trained models fire_detection_model.h5 and stray_animal.h5 for image classification.

```
1. from flask import Flask, request, jsonify
2. from PIL import Image
3. import numpy as np
4. import tensorflow as tf
5. import nltk
6. from nltk.tokenize import word_tokenize
7.
8. app = Flask(__name__)
9.
10. # Load custom trained CNN models
11. model = tf.keras.applications.load_model(weights='fire_detection_model.h5')
12.
13. # Hazard keyword mapping
14. hazard_keywords = {
15.     "fire": ["fire", "burn", "smoke", "flames"],
16.     "flood": ["flooding", "flood", "water", "rain", "overflow"],
17.     "accident": ["accident", "crash", "collision"]
18. }
19.
20. # NLP FUNCTION — NLTK keyword-based classification
21. def analyze_text(text):
22.     tokens = word_tokenize(text.lower())
23.     for label, words in hazard_keywords.items():
24.         match_count = sum(1 for word in tokens if word in words)
25.         if match_count > 0:
26.             confidence = min(0.5 + (match_count * 0.1), 0.9)
27.             return label, confidence
28.     return "unknown", 0.2
29.
30. # CNN FUNCTION — Custom model image classification
31. def analyze_image(img_path):
32.     img = Image.open(img_path).resize((224, 224))
33.     img_array = np.array(img)
34.     img_array = np.expand_dims(img_array, axis=0)
35.     img_array = tf.keras.applications.preprocessing.image\
36.         .preprocess_input(img_array)
37.     preds = model.predict(img_array)
38.     decoded = tf.keras.applications.preprocessing.image\
39.         .decode_predictions(preds, top=1)[0][0]
40.     label = decoded[1].lower()
41.     confidence = float(decoded[2])
42.     if "fire" in label or "smoke" in label: return "fire", confidence
43.     elif "water" in label or "flood" in label: return "flood", confidence
44.     elif "car" in label or "truck" in label: return "accident", confidence
45.     else: return "unknown", confidence
46.
47. @app.route('/analyze', methods=['POST'])
48. def analyze():
49.     text = request.form['text']
50.     image = request.files['image']
```

```

51.     path = "temp.jpg"
52.     image.save(path)
53.     text_label, text_conf = analyze_text(text)
54.     img_label, img_conf = analyze_image(path)
55.
56.     # Trust score computation
57.     if text_label == img_label and text_label != "unknown":
58.         trust_score = (text_conf * 50) + (img_conf * 50)
59.         status = "REAL"
60.     else:
61.         trust_score = (img_conf * 20)
62.         status = "SUSPICIOUS"
63.
64.     return jsonify({
65.         "text_label": text_label,
66.         "text_conf": text_conf,
67.         "image_label": img_label,
68.         "image_conf": img_conf,
69.         "trust_score": trust_score,
70.         "status": status
71.     })
72.
73. if __name__ == '__main__':
74.     app.run(debug=True, port=5000)
75.

```

APPENDIX 3 — DATABASE SQL QUERIES

The following SQL queries define the two database tables used in the NeighbourGuard system.

```

1.  -- USERS & ADMINS TABLE
2.  CREATE TABLE users (
3.      id          INT AUTO_INCREMENT PRIMARY KEY,
4.      name        VARCHAR(100),
5.      email       VARCHAR(150) UNIQUE,
6.      password    VARCHAR(255),
7.      google_id   VARCHAR(255),
8.      login_type  VARCHAR(20) DEFAULT 'manual',
9.      role        ENUM('user', 'admin') DEFAULT 'user',
10.     otp          VARCHAR(10),
11.     created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
12. );
13.
14. -- REPORTS TABLE
15. CREATE TABLE reports (
16.     id          INT AUTO_INCREMENT PRIMARY KEY,
17.     name        VARCHAR(100),
18.     user_id     INT,
19.     description TEXT,
20.     latitude    DECIMAL(10,8),
21.     longitude   DECIMAL(11,8),
22.     image       VARCHAR(255),
23.     audio       VARCHAR(255),
24.     hazard_type VARCHAR(255),
25.     alert_message VARCHAR(255),
26.     report_time  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
27. );
28.

```

APPENDIX 4 — TRUST SCORE DECISION LOGIC

The following pseudocode describes the complete trust score computation and storage decision logic.

```
1. // PHP Random Forest Score
2. text_length_score = (len(description) >= 20) ? 25 : 0
3. keyword_score     = (no fake words in description) ? 25 : 0
4. image_score       = (image uploaded) ? 25 : 0
5. audio_score       = (audio uploaded) ? 25 : 0
6. rf_score          = text_length_score + keyword_score + image_score + audio_score
7.
8. // Flask AI Score (from NLP + CNN analysis)
9. IF text_label == image_label AND text_label != 'unknown':
10.     ai_trust_score = (text_confidence * 50) + (image_confidence * 50)
11.     status = 'REAL'
12. ELSE:
13.     ai_trust_score = (image_confidence * 20)
14.     status = 'SUSPICIOUS'
15.
16. // Final Combined Trust Score
17. final_trust_score = (rf_score * 0.5) + (ai_trust_score * 0.5)
18.
19. // Storage Decision
20. IF final_trust_score > 60:
21.     INSERT INTO reports table // Report ACCEPTED
22. ELSE:
23.     // Report REJECTED – Nothing stored in database
24.
25.
```

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